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Persistent Dealer-Gamma Regimes Under Temporal Obfuscation: A Re-Run on Complete Inputs with a Controlled LLM Re-Test

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Abstract: Temporal obfuscation tests whether a large language model (LLM) reasons about numbers or recalls market history. Our earlier work reported that an LLM detects persistent dealer-gamma regimes in 30-day windows of obfuscated daily gamma exposure (GEX) of SPY, with a sharp rise after 2023. We re-run that test on the complete data. The published 2024–2025 inputs lacked the put leg of the options chain. We assembled six daily series (recovered, archived or rebuilt), replayed the deterministic rule, bridged o4-mini to gpt-6-luna on the published inputs, reran the 2024 and 2020 baselines with relative day labels, and repeated seven arms with an open-weights model. On the published inputs the rule detects all 501 windows of 2024–2025; on the complete series it detects 60 of 502 (12.0%), and gpt-6-luna 63 of 502 (12.5%). The two models agree on 99.6% of the published windows (Cohen's kappa 0.993), and gpt-6-luna follows the rule in 99.7% of 4,691 windows (kappa 0.985); gpt-oss-20b agrees with gpt-6-luna on 99.4%. Removing same-day expiries gives 14.7%, not the published rate. With relative labels gpt-6-luna detects 80.7% of the 2024 baseline windows, against 81.2% with dates. The complete series shows no post-2023 increase that the data distinguish from zero.

Keywords: LLM validation; regime detection; gamma exposure; 0DTE options; temporal obfuscation; data provenance; reproducibility

1. Introduction

A language model that labels a market regime may be recalling the period it was trained on, or it may be reasoning about the numbers it is shown. Temporal obfuscation separates the two readings: dates and tickers are removed, so that the model has only the values and the stated criteria. The test is only as sound as the numerical series that it shows to the model. This article re-runs the published 30-day regime test on the complete daily series, with a second model on identical inputs, and asks what the published conclusions rest on.

1.1. Motivation: Data-Definition Drift and Model Retirement

The published run used o4-mini in January 2026. The model no longer answers inference requests (the endpoint returns `model_not_found`). The raw options table was reloaded on 2026-02-03, and the daily GEX table was rebuilt on 2026-10-08 from stored per-contract gamma, with same-day expiries included (Section 3.3). A result that depends on a model and on an input definition that can both disappear cannot be re-run, only re-interpreted. The study therefore recovers the exact inputs of the

published run from the stored batch file, bridges a second model (gpt-6-luna) to o4-mini on these inputs, and stores every input, prompt and response. Because a hosted alias can be retired as well, seven arms are repeated with an open-weights model at a fixed revision that runs on local hardware.

1.2. Relation to Prior Work

This article is the extended version of the AIAI 2026 conference paper [1], which introduced 30-day regime detection under temporal obfuscation and built on the single-day study presented at IEEE BigData 2025 [2]. Both lines of work are combined in a separate journal article [3] (Journal of Risk and Financial Management, JRFM).

Relative to all three publications, the new elements are:

(i) the provenance and definition of the daily input series;
(ii) a second model, gpt-6-luna, bridged to o4-mini on identical inputs;
(iii) the years 2008–2019, the exchange-traded funds (ETFs) SPY (S&P 500), QQQ (Nasdaq-100) and IWM (Russell 2000), 2026 and an intraday check;
(iv) overlap-aware inference;
(v) an obfuscated-versus-dated rerun of the Phase 3 and 4 baselines;
(vi) a disclosed per-symbol scaling for QQQ and IWM; and
(vii) a third model with open weights, gpt-oss-20b, run locally at a fixed revision on seven arms.

The recap of the earlier design is limited to Section 3.

1.3. Research Questions

We address eight questions:

RQ1. Which daily series were the published inputs, and how do the deterministic labels change on the complete series?
RQ2. Does gpt-6-luna reproduce the o4-mini labels on the exact published inputs?
RQ3. Does the model's regime call follow the input series and the stated criteria?
RQ4. How much of the persistence signal lives in zero-days-to-expiration (0DTE), i.e. same-day-expiry, gamma?
RQ5. Do the answers hold for 2008–2019, for QQQ and IWM, and in the post-publication period (2026)?
RQ6. Does a late-session intraday snapshot reproduce the end-of-day label?
RQ7. Does the 2024-versus-2020 contrast of the earlier baseline runs, whose prompts carried calendar dates, persist when the dates are replaced by relative day labels?
RQ8. Does an open-weights model, run locally at a fixed revision, give the verdicts of gpt-6-luna and of the rule?

RQ1 is answered in Sections 4.1–4.2, RQ2 in Section 4.3, RQ3 in Sections 4.4 and 4.10, RQ4 in Section 4.5, RQ5 in Sections 4.6–4.8, RQ6 in Section 4.9, RQ7 in Section 4.11 and RQ8 in Section 4.12.

1.4. Contributions

This article makes seven contributions. (i) It reconstructs the published inputs and recomputes the deterministic labels on the complete series (Sections 4.1 and 4.2). (ii) It analyses the sensitivity of the deterministic labels to the series definition across six daily series (one of them a call-only diagnostic), three assets and 2008–2026. Removing same-day expiries does not restore the published rates, and the fixed thresholds leave QQQ and IWM without a detection until a disclosed scaling is applied (Sections 4.5–4.7). (iii) It bridges o4-mini and gpt-6-luna on identical inputs. The two models give nearly the same verdict on the published windows, and on the complete series gpt-6-luna follows the rule almost everywhere (Sections 4.3, 4.4 and 4.10). (iv) It applies overlap-aware statistics to stride-one windows: the quarter-block bootstrap and a non-overlapping subsample next to the Wilson intervals. Wilson and bootstrap intervals are reported for the pooled rule rate (Section 3.8 for the

method, Section 4.2 for the values). (v) It stores every input, prompt and response in result tables (49,090 answers), from which each table and figure of this article is generated (Section 3.7). (vi) It reruns the Phase 3 and 4 baselines with relative day labels instead of calendar dates. For 2024, gpt-6-luna detects 180 of 223 windows (80.7%) with relative labels and 181 of 223 (81.2%) with dates, so the visible dates changed nothing that the data can detect (Section 4.11). (vii) It repeats seven arms with an open-weights model, gpt-oss-20b, at a fixed revision on local GPUs. The model agrees with gpt-6-luna in 99.4% of the windows of the complete series and reproduces the bridge and the absence of a date effect, so these results do not depend on the hosted alias (Section 4.12).

1.5. Organization

Section 2 reviews dealer gamma exposure, the validation of LLM reasoning by obfuscation and the construct validity of financial inputs. Section 3 defines the daily series, the windows and rule, the model protocol, the experimental arms and the statistics. Section 4 reports the results, from the lineage of the published inputs to the rerun of the earlier baselines and the open-weights model. Section 5 discusses what survives, what the model tracks, the threats to validity and the implications. Section 6 concludes, and the appendices give the prompt, schema and API configuration (Appendix A), the per-year tables (Appendix B) and the data availability details (Appendix C). The series definitions in SQL and the run ledger are in the Supplementary Materials.

2. Background and Related Work

2.1. Dealer Gamma Exposure and 0DTE Options

Market makers who hedge their books toward delta-neutral turn accumulated gamma into systematic hedging flows [4,5]. Dealers are the counterparties of customer option positions, and aggregated market makers carry negative gamma that feeds back to the spot market [6]. Dealer hedging has measurable price effects [7–9], and practitioners summarise it as a gamma exposure (GEX) figure [6,10,11]. Such a figure is a lossy scalar: it depends on modelling choices (which gamma, which contracts, which expiries), so two daily GEX series with the same name need not agree. It also rests on an assumption about positions: open interest records how many contracts are open but not which side dealers hold, so the convention used here (Equation (1)) assigns dealers a long position in calls and a short position in puts.

The growth of zero-days-to-expiration (0DTE) options since 2022 concentrates gamma at very short maturities [12–15]. Whether 0DTE contracts are included is one such modelling choice, and it can change the sign of the daily figure on some days. Regime-switching models detect regimes from observable outcomes such as returns [16–18]. The rule and the LLM in this paper work on dealer positioning instead.

2.2. Validating LLM Reasoning via Obfuscation

LLMs are applied to financial tasks such as return prediction from news [19] and multi-agent trading [20], and chain-of-thought prompting makes their reasoning steps explicit [21,22]. A visible reasoning trace does not show where an answer came from. Concern that LLM performance reflects memorized rather than inferred structure [23–25] motivates behavioral tests of the CheckList type [26]. Temporal obfuscation removes dates, tickers and event context so that only numerical structure remains; it is one such test [2]. A model that still gives the same answer on obfuscated input is less likely to be recalling training data [24].

Obfuscation has a limited reach. It tests whether the answer depends on memorized context. It does not test whether the numbers shown to the model mean what the prompt says they mean. A model can pass an obfuscation test and still report faithfully on an input whose definition differs from what the prompt states. That second question concerns the inputs, not the model, and is the subject of the next subsection.

2.3. Construct Validity of Financial Inputs

An LLM validation inherits the construct validity of its inputs [27]. A daily GEX series is a derived quantity. Its definition (gamma model, contract filters, 0DTE treatment, put and call legs) and its provenance (vendor, load date, rebuild date) are part of the experiment, and they belong in the record of a study as much as the prompt and the model name do.

Three threats follow for the reproducibility of LLM studies. Data definitions drift between collections. Upstream data can be rebuilt silently, so that the same query returns a different series later. Models are deprecated, so a run cannot be repeated with the same model. Section 3 describes how this paper records the input series and the model version for every arm.

3. Methods

The study cuts six daily GEX series and the derived inputs of the rerun arms (39,027 stored daily rows) into 38,592 windows of 30 trading days. Each window is labelled by the deterministic rule and, for each of 25 arms (15 gpt-6-luna arms, seven gpt-oss-20b arms and three o4-mini backfills; the two 40-request pilots are listed beside them in Table 3), by a language model; the 49,090 stored answers (1,922 of them backfilled from earlier o4-mini runs and 12,384 from gpt-oss-20b) and all other results are kept in parquet files and in the tables `backups.gex_llm_regime_*`. Figure 1 gives an overview. The daily series are defined in Section 3.2 and their lineage in Section 3.3; windows and the rule follow in Section 3.4, the model protocol in Section 3.5 and the experimental arms in Section 3.6.

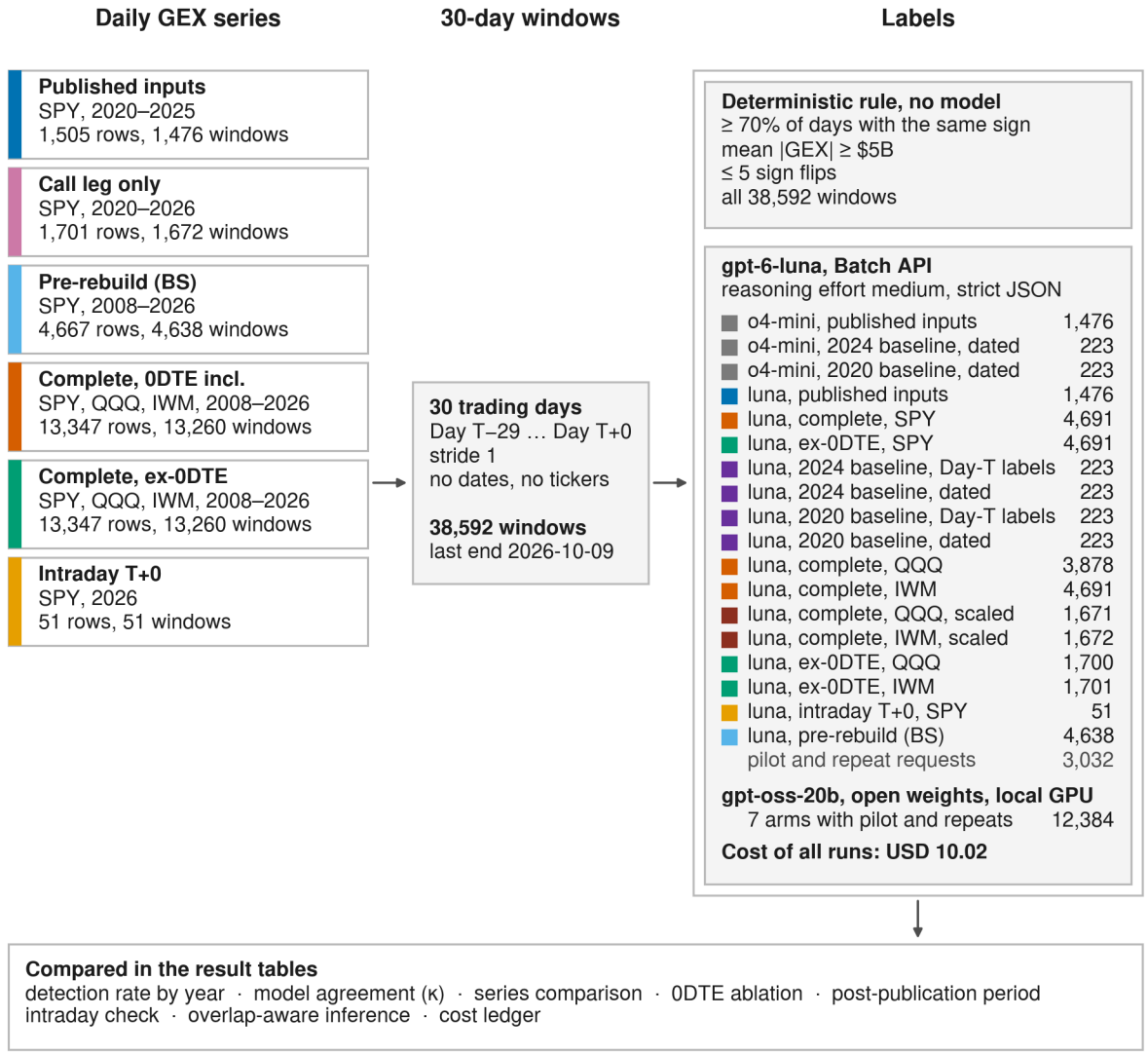


Figure 1. Study design. Six daily GEX series, one of them the call-only diagnostic, and the derived inputs of the rerun arms (Table 1) are cut into 30-trading-day windows, labelled by the deterministic classifier and, for each arm of Table 3, by gpt-6-luna through the Batch API or, for the seven open-weights arms, by gpt-oss-20b on a local GPU. Boxes give the stored counts, and every input, label and response is stored (Section 3.7). **Read this figure as:** daily rows become 38,592 windows, every window carries a label from the rule, and each of the 49,090 stored answers is compared with the label of its own window.

3.1. Regime Criteria

A 30-trading-day window is a persistent regime when three criteria hold [1,3]: (i) persistence, at least 21 of the 30 days (70%) share the sign of net GEX; (ii) magnitude, the mean absolute net GEX is at least $5 billion; and (iii) stability, the sign changes at most five times. The deterministic classifier (the rule) labels every other window as low conviction (a persistent sign as in (i) and a mean magnitude of at least $3 billion, but failing (ii) or (iii)) or transitional. The detection label used throughout is the binary persistent or not-persistent verdict (`regime_detected` for the model, `clf_should_detect` for the rule). The thresholds are those of the earlier studies and are not re-tuned. Zero GEX counts as non-positive. A window with a persistent sign and a mean magnitude of at least $5 billion but more than five sign changes fails the stability criterion and is labelled low conviction, because its mean still exceeds $3 billion. The prompt shows values rounded to 0.01B, whereas the rule replays the

unrounded USD values stored with each window. On 4 of the 38,592 windows the rounded values would give another four-way label, and on two of them (scaled QQQ, window ends 2021-03-05 and 2021-03-08) another verdict. The thresholds are fixed in dollars before the runs, for every symbol. This choice has a consequence for QQQ and IWM, whose gamma is smaller than that of SPY, and Section 4.7 reports the resulting floor effect.

3.2. Daily GEX Series

For each symbol and trading day t the net dealer gamma exposure is

$$\mathrm{GEX}_t = \sum_{i \in \mathcal{C}_t} s_i \, \Gamma_i \, \mathrm{OI}_i \cdot 100 \cdot S_t^2 \cdot 0.01, \qquad s_i = \begin{cases} +1 & \text{call}, \\ -1 & \text{put}, \end{cases} \tag{1}$$

where $\mathcal{C}_t$ is the set of contracts listed on day t with a defined gamma Γ_i, positive open interest OI_i and a positive underlying price S_t. The series of Table 1 differ in how Γ_i is obtained, in which expiries and legs enter $\mathcal{C}_t$, and in where the values come from.

Table 1. Daily GEX series. The first block holds the series of the published result and its diagnostics, the second the complete series, the third the intraday check, the fourth the derived inputs of the rerun arms (Section 3.6).

Series key	Source and gamma convention	Symbols and span	Role
`jan2026_phase4a`	Phase 4A batch file of 2026-01-04: the exact inputs sent to o4-mini, at 0.01B precision	SPY, 2020-01-02 to 2025-12-31	Bridge to the published result
`call_only`	Raw `gex.options_chains_partitioned`: call leg of every contract, same-day expiries included, put leg dropped	SPY, 2020 to 2026-10-09	Reproduces the put-less inputs; no model arm
`old_bs`	Archive `backups.options_daily_summary_pre_rebuild_20261004`: Black–Scholes gamma on integer day counts, same-day expiries zeroed; put-less from about October 2023	SPY, 2008-01-02 to 2026-10-02, with holes	Diagnostic bridge to the pre-rebuild table
`new_full`	`gex.options_daily_summary` after the 2026-10-08 rebuild and the additions of 2026-11-03 (Section 3.3): stored per-contract gamma, same-day expiries included, both legs	SPY, QQQ, IWM; 2008 (QQQ 2011) to 2026-10-09	Primary complete series
`new_ex0dte`	Raw `gex.options_chains_partitioned`, contracts with `expiration > trading_date`; otherwise as `new_full`	As `new_full`	0DTE ablation
`intraday_late`	`gex.intraday_gex_summary`, snapshot type `final_rush` (19:55:59 UTC), stored gamma	SPY, 2026-07-27 to 2026-10-09	Intraday check
`new_full_scaled`	`new_full` of QQQ and IWM times one fixed factor per symbol (Section 3.6)	QQQ, IWM; 2020 to 2026-10-09	Scaled cross-asset arms (protocol change)
`p3_2024`	Phase 3 batch file of 2025-11-20: 252 daily values, gross call-plus-put magnitude without same-day expiries and with a negative sign imposed (negative on 242 days; the ten positive values are placeholder rows of the source database)	SPY, 2024-01-02 to 2024-12-31	2024 baseline, Day-T labels
`p3_2024_dated`	The same values with the ISO dates that were sent in November 2025	As `p3_2024`	2024 baseline, dated labels
`p4_2020`	Phase 4 batch file of 2025-11-20: 252 daily values, magnitude only (positive on every day)	SPY, 2020-01-02 to 2020-12-31	2020 baseline, Day-T labels
`p4_2020_dated`	The same values with the ISO dates that were sent in November 2025	As `p4_2020`	2020 baseline, dated labels

Each series answers one question. `jan2026_phase4a` holds the numbers that o4-mini saw in January 2026, so it is the only input that can link the new runs to the published rates. `call_only` is rebuilt from the raw chain with the call leg alone and tests whether the put-less inputs alone explain the published rates. `old_bs` is the pre-rebuild daily table and shows how far a Black–Scholes [28] gamma on integer day counts moves the rates. `new_full` is the complete series and the primary object of

the study. `new_ex0dte` removes same-day expiries and so isolates their contribution. `intraday_late` supplies a late-session value for the last day of a window. The fourth block serves the cross-asset and baseline replays. `new_full_scaled` rescales QQQ and IWM by one factor per symbol (Section 3.6), so that the fixed dollar thresholds are comparable across the three ETFs. The Phase 3 file carries a negative sign on 242 of its 252 days (the ten positive values are placeholder rows), the Phase 4 file is positive on all 252 days, and the dated twins differ from their obfuscated counterparts in the day labels only.

Windows use the rows of each series (rule A, Section 3.4), so the calendars matter. The series `old_bs` lacks 53 trading days of the complete series for SPY: the 19 days of 2009 to 2018 and the day 2026-05-06 that were added to the daily table on 2026-11-03 (Section 3.3), 2023-07-12, 2024-03-11, the 26 trading days from 2026-01-02 to 2026-02-09, and the 5 days from 2026-10-05 to 2026-10-09. The January calendar lacks 2023-07-12 and 2024-03-11; both dates are in the complete series. For the intraday series, the `market_close` slot (20:00:59 UTC) equals the end-of-day row on 51 of 54 days, so the independent late-session observation is `final_rush`. The irregular days 2026-07-29, 2026-08-03 and 2026-09-14 are excluded.

3.3. Provenance of the Published Inputs

The published inputs cannot be regenerated from today's database. The raw options table was reloaded on 2026-02-03, one month after the Phase 4A run, and a defect of the earlier load (option type defaulting to call) had removed the put rows of the raw table for the dates from about October 2023 to December 2025. The exact inputs were therefore recovered from the surviving Phase 4A batch file (Section 4.1); the inputs of the Phase 3 and 4 baseline runs of November 2025 survive in the same way. Nine dated events fix the lineage:

- 2025-11-20: the Phase 3 (2024) and Phase 4 (2020) baseline batches are run on o4-mini with calendar dates as day labels, 223 windows each (Section 3.6).
- 2026-01-04: Phase 4A batch run (o4-mini-2025-04-16) on 1,476 SPY windows with ends from 2020-02-13 to 2025-12-31.
- 2026-01-15: the pre-rebuild daily table is computed from the same raw table with Black–Scholes gamma and same-day expiries zeroed.
- 2026-02-03: the raw options table is reloaded, with the put rows restored.
- 2026-09-14: the model object gpt-6-luna is created.
- 2026-10-04: the pre-rebuild daily table is archived and treated as read-only.
- 2026-10-08: the daily summary is rebuilt from stored per-contract gamma, same-day expiries included.
- 2026-10-12: equity options move from Alpha Vantage to Cboe delayed quotes, so every window ends on or before 2026-10-09.
- 2026-11-03: the database team adds daily rows to `gex.options_daily_summary` for SPY, QQQ and IWM that had been missing. For 2026-05-06 the chain was absent that day and was recovered from the intraday slot of the next morning, which carries the previous session's chain. For 41 symbol-days of 2008 to 2018 (SPY 19, QQQ 10, IWM 12) the stored chain lacked the spot price; for these the summary uses the close of the day, accepted only where the put-call parity of the chain puts the spot within 1% of it (largest deviation 0.35%). The days are listed in `gex.spot_fill_log` and our series mark them in `series_metadata` (`spot_from`). The series built from the raw chain take the same spot on exactly these days and equal the summary to within 2.6×10^{-7} relative. Four days stay missing because no chain exists for them: 2018-11-06, 2019-09-03 and 2020-06-23 for all three symbols, and 2024-09-30 for QQQ.

Two observations carry the evidence on the put-row loss; the others, including the share of days on which the January value equals the complete series in each year, are in Table 4 and Section 4.1. First, on each of the 565 trading days from 2023-09-29 to 2025-12-31 the January value equals the call-only GEX of today's raw table, same-day expiries included, to within \$0.005 billion, and it equals the complete

net GEX on none of them. Second, the January calendar lacks 2023-07-12 and 2024-03-11, two trading days that the reloaded table holds; the 58 January windows that span them are flagged (Section 3.4). The cause (option type defaulting to call) is recorded in issue 24 of the database repository.

The pre-rebuild table is therefore not a clean old convention. The effect of Black–Scholes against stored gamma is identifiable only before about October 2023, and the effect of same-day expiries is the difference between `new_full` and `new_ex0dte`.

3.4. Windows and Deterministic Classifier

A window consists of 30 consecutive rows of a series (days $T-29$ to $T+0$), with one window per end date (stride one) and end dates up to 2026-10-09. A span that covers a calendar gap of more than four days is flagged (a gap of five or more days is not a holiday weekend). The exchange closure of 2012-10-29 and 2012-10-30 (Hurricane Sandy) is such a gap without a missing trading day, so in each series that covers 2012 (`new_full` and `new_ex0dte` for the three symbols, and `old_bs`) the 29 windows that span it are flagged although they are complete. The number of trading days of the symbol's master calendar (`gex.options_daily_summary`) that are absent from the series inside the span is stored with the window. A third detector compares the span with the NYSE trading calendar and stores the number of NYSE trading days that the series lacks inside it (`nyse_missing_days`); it finds a single day that is absent from the series and from the master calendar alike, which the first two detectors cannot see. A flag marks a window but does not drop it, and flagged windows are kept in every rate. The gap detector fires on two spans of the complete series, the closure of 2012-10-29 and 2012-10-30, when no day is missing, and the absence of 2019-09-03; the NYSE detector fires on 2018-11-06, 2019-09-03 and 2020-06-23 for all three symbols and on 2024-09-30 for QQQ. Values are relabelled "Day T-29" to "Day T+0" with no dates or tickers, exactly as in Phase 4A, except in the two dated arms of Section 3.6, which carry the ISO dates of the November 2025 prompts. The rule of Section 3.1 labels every window independently of any model call, and its verdict is the reference label for each arm. The windows that end in 2026, up to 2026-10-09, form the post-publication period (Section 4.8): the published sample ends on 2025-12-31, so the 29 earliest of these windows still contain days of it, and those with ends from 2026-02-13 lie entirely after it. Table 2 gives the 38,592 windows. The January inputs and the four Phase 3 and 4 inputs keep their own calendars (rule J), and a build stops unless it reproduces the surviving batch file byte for byte. The composite intraday windows (rule T0, key `intraday_t0`) take the `new_full` values for days $T-29$ to $T-1$ and the `final_rush` value at $T+0$. A day without a finite value is left out of a rebuilt series before the windows are cut, and stops the window build for the recovered January and Phase 3 and 4 inputs, which must stay as they were sent; no stored series has such a day, so no window holds a missing value. Of the 1,476 January windows, 58 span one of the two missing dates and are flagged. Windows overlap in 29 of 30 days, which the statistics of Section 3.8 take into account.

Table 2. Windows per series and symbol. Rule A cuts 30 consecutive rows of a series that follows the master calendar, rule J does the same for a series with its own calendar, and rule T0 builds the composite intraday windows. The last four columns count windows whose widest calendar gap exceeds four days (Gap), windows with at least one master-calendar day absent from the series (Master), windows with at least one NYSE trading day absent from the series (NYSE), and windows with any of the three flags (Either). Flagged windows are kept.

					Windows flagged by			
Series key	Symbol	Rule	Windows	Window ends	Gap > 4 d	Master	NYSE	Either
`jan2026_phase4a`	SPY	J	1,476	2020-02-13 to 2025-12-31	0	58	87	87
`call_only`	SPY	A	1,672	2020-02-13 to 2026-10-09	0	0	29	29
`old_bs`	SPY	A	4,638	2008-02-13 to 2026-10-02	116	389	476	505
`new_full`	IWM	A	4,691	2008-02-13 to 2026-10-09	58	0	87	116
`new_full`	QQQ	A	3,878	2011-05-04 to 2026-10-09	58	0	116	145
`new_full`	SPY	A	4,691	2008-02-13 to 2026-10-09	58	0	87	116
`new_ex0dte`	IWM	A	4,691	2008-02-13 to 2026-10-09	58	0	87	116
`new_ex0dte`	QQQ	A	3,878	2011-05-04 to 2026-10-09	58	0	116	145
`new_ex0dte`	SPY	A	4,691	2008-02-13 to 2026-10-09	58	0	87	116
`intraday_t0`	SPY	T0	51	2026-07-27 to 2026-10-09	0	0	0	0
`new_full_scaled`	IWM	A	1,672	2020-02-13 to 2026-10-09	0	0	29	29
`new_full_scaled`	QQQ	A	1,671	2020-02-13 to 2026-10-09	0	0	58	58
`p3_2024`	SPY	J	223	2024-02-13 to 2024-12-31	0	0	0	0
`p3_2024_dated`	SPY	J	223	2024-02-13 to 2024-12-31	0	0	0	0
`p4_2020`	SPY	J	223	2020-02-13 to 2020-12-31	0	0	29	29
`p4_2020_dated`	SPY	J	223	2020-02-13 to 2020-12-31	0	0	29	29
Total			38,592		464	447	1,317	1,520

3.5. Model Protocol

The main model is OpenAI's `gpt-6-luna`, called through the Batch API (`/v1/chat/completions`, 24-hour completion window) in chunks of at most 1,500 requests. The model replaces the retired o4-mini; its API alias carries no dated snapshot, so the creation timestamp of the model object (2026-09-14) and the response identifier of every answer are stored. Requests use `reasoning_effort` = `medium`, the effort at which o4-mini ran in the published run, and the prompt text is unchanged from the Phase 4A run (prompt version `v2.0_postgresql`); only the two dated arms of Section 3.6 differ in the data lines, whose day labels are ISO dates. The wording of the prompt uses "$>$70%" and "$>$\$5B", whereas the rule uses "at least 21 of 30 days" and "$\geq$ \$5 billion"; the prompt is unchanged from the published run. The answer has nine fields (Appendix A). The one protocol change that bears on the answer is that these fields are enforced with a strict JSON schema, whereas the earlier runs requested JSON in the prompt text only; the change is disclosed here and does not alter the question asked. The published requests supplied neither a temperature nor a token limit (the AIAI Methods text states 1.0 and 16,384 tokens); the requests to gpt-6-luna supply no temperature either. They do cap completion at 4,000 tokens, reasoning tokens included, as a safety limit that no answer reached. An answer is read as JSON; a second attempt tolerates raw control characters inside strings (one Phase 4A answer needed it), a last fallback extracts the JSON object from free text, and the parse status of every answer is stored.

A pilot of 40 windows, balanced on the label of the rule (20 detected, 20 not) and run synchronously at standard prices, measured length and cost before the Batch runs. All 40 answers parsed on the first strategy and all 40 verdicts equal the rule and the published o4-mini answer. A request used 1,913 prompt tokens and on average 700 completion tokens, of which 513 were reasoning tokens; the longest answer used 1,063 completion tokens. The pilot cost \$0.02 in all. Over all gpt-6-luna runs, the pilot included, the study sent 35,833 requests, 1,121 of them for windows re-requested after the rebuild of 2026-11-03 (Section 3.7); the 34,784 gpt-6-luna answers that are stored parsed (100.0%), 0 needed the fallback parsers and 0 ended with `finish_reason` = `length`, so the retry at 8,000 tokens was never used. A stored answer averaged 1,914 prompt and 734 completion tokens, and the longest answer

used 1,864. At Batch prices of $0.05 and $0.25 per million input and output tokens, 1,000 requests cost $0.28.

The bridge carries a gate. Cohen's kappa between `bridge_jan` and the 1,412 published o4-mini answers has to be at least about 0.6, the value of the project checklist. The observed kappa is 0.99 (99.6% agreement: 646 windows detected by both models, 5 by gpt-6-luna only, 0 by o4-mini only, 761 by neither), so the gate is passed. A full repeat of `bridge_jan` gives a kappa of 0.98 against the first pass (1,476 pairs), which sets the scale of the run-to-run noise.

A third model answers seven arms (Section 3.6). The alias gpt-6-luna can be retired as o4-mini was, so these arms use a model whose weights can be run again: gpt-oss-20b [29], an open-weights model released by OpenAI under the Apache-2.0 licence, at the fixed revision `6cee5e81` of its public repository `openai/gpt-oss-20b`. It runs on the authors' NVIDIA A100 GPUs with the offline batch runner of vLLM 0.32.0 [30], which reads request files in the format of the Batch API, and the weights are read from a local copy of that revision without network access. The request body is that of gpt-6-luna with only the model name changed: the same prompt, `reasoning_effort` = `medium`, the strict JSON schema and the limit of 4,000 completion tokens, with one more request at 8,000 tokens for an answer cut off at the limit. vLLM applies the schema through its structured-output decoding once the reasoning channel of the model has ended. No temperature is sent, so the engine samples at its default of 1.0, with the engine seed fixed at 261; answers still need not repeat exactly, because requests share the engine in batches whose composition varies. The engine arguments, the vLLM version, the weights revision, the GPU and the run time are stored with every run. These runs are `local` in the run ledger: they use no API and cost nothing beyond the GPU time, so they do not enter the cost of the study.

gpt-oss-20b comes from the developer of gpt-6-luna, so a fourth model answers seven more arms: Llama 3.1 8B Instruct [31], released by Meta under the Llama 3.1 Community License, at the fixed revision `0e9e39f2` of `meta-llama/Llama-3.1-8B-Instruct`, in its release precision (bfloat16). It runs in the same way as gpt-oss-20b, on the same engine. Llama 3.1 has no reasoning mode, so its requests carry no `reasoning_effort`; the body is otherwise unchanged. The schema lists the verdict first and the reasoning last, and the structured-output decoding follows that order, so this model writes its verdict before any reasoning, whereas the three reasoning models reason before they write the JSON object. Sampling follows the defaults of the repository (temperature 0.6, top-p 0.9), with the same engine seed.

Llama 3.1 8B has no reasoning phase, so a fifth model, from a third developer, answers the same seven arms: Qwen3-8B [32], released by the Qwen team of Alibaba under the Apache-2.0 licence, at the fixed revision `b968826d` of `Qwen/Qwen3-8B`, in bfloat16 and in its default thinking mode. It runs on the same engine with the reasoning parser of Qwen3, so that, as for gpt-oss-20b, the schema applies once the reasoning has ended. The request body is that of gpt-6-luna with the model name changed; the chat template of Qwen3 does not use `reasoning_effort`. Sampling follows the defaults of the repository (temperature 0.6, top-p 0.95, top-k 20), with the same engine seed. The engine took 27% of the GPU memory, or up to 45% when the shared GPUs had more free, which sets the cache and batch size but not the model or the sampling. Qwen3-8B reasons at length (about 7,000 completion tokens per window in its pilot), so many answers reach the limit of 4,000 tokens, and some the retry limit of 8,000, before the JSON object is written. Its 40-window pilot parsed 36 answers, short of the 95% that the protocol required of every pilot; before the arms ran, the authors amended the protocol to run them unchanged and to report a window whose answer does not finish within 8,000 tokens as unparsed (Section 4.12).

3.6. Experimental Arms

Table 3 lists the arms. The arm `r1_o4mini` holds the published o4-mini answers, backfilled from the surviving Phase 4A output file without an API call; 64 of its 1,476 answers were lost to a parsing error in January 2026 and are re-parsed from the stored raw output. Every headline number uses the 1,412 answers of the published result; the 64 re-parsed answers are reported separately (as

`r1_o4mini_all`). The arms `r1_p3_o4mini` and `r1_p4_o4mini` are backfilled in the same way from the 223 stored answers of each November 2025 baseline run. The arm `bridge_jan` sends the exact published inputs to gpt-6-luna, which separates the effect of the model from the effect of the inputs. The other arms run gpt-6-luna on the complete series of each symbol; the ex-0DTE arms of QQQ and IWM start in 2020. The intraday arm uses composite windows whose last day is the late-session snapshot.

Two further groups of arms test the earlier results more directly. The Phase 3 (2024) and Phase 4 (2020) baselines of the earlier studies [1,3], the 81.2% versus 12.1% contrast, were run on o4-mini in November 2025 with ISO calendar dates as day labels instead of relative labels. The arms `p34_dated_2024` and `p34_dated_2020` send the same 252 values per year, with the dates that were actually sent, to gpt-6-luna; the arms `p34_obf_2024` and `p34_obf_2020` send the identical values with Day-T labels. Within a year the two arms differ in the labels only, which isolates the effect of visible dates, and each is compared with the stored o4-mini answers. The two baselines are not on one convention (Table 1): the 2024 series is a gross call-plus-put magnitude with a negative sign imposed and the 2020 series is a magnitude without sign. We keep them verbatim because the aim is to replay the published contrast, which therefore cannot be read as one between a negative and a positive regime.

With the thresholds fixed in dollars, QQQ and IWM have no persistent window (Section 4.7), so the unscaled arms only test whether the model also rejects every window. As a disclosed protocol change, the arms `full_qqq_scaled` and `full_iwm_scaled` therefore run on `new_full_scaled`: the `new_full` values of QQQ and IWM multiplied by one fixed factor per symbol, the median over 2020-01-01 to 2026-10-09 of SPY's open-interest notional divided by the symbol's (`total_weighted_oi` $\times$ 100 $\times$ underlying price, from `gex.options_daily_summary`; Section S1 of the Supplementary Materials). The factors are 4.22 for QQQ (over 1,699 days) and 11.43 for IWM (over 1,700 days), the values of the build of 2026-10-29, and they are stored with the series. They were kept at these values after the additions of 2026-11-03 because the scaled arms had been run on them; the medians recomputed with the added days (1,700 and 1,701 days) are 4.22 and 11.42 and are recorded in the series metadata. Prompt, thresholds and rule are unchanged, so scaled and unscaled arms are labelled by the same rule and run side by side.

The open-weights arms carry the prefix `oss_` for gpt-oss-20b, `llama_` for Llama 3.1 8B and `qwen_` for Qwen3-8B: each sends the requests of the gpt-6-luna arm of the same name, on the same windows, to that model (Section 3.5). The three sets cover the bridge, the two complete SPY arms and the four baseline arms, which are the arms that bear on the published result and on the effect of dates; the QQQ, IWM, intraday and archive arms are not repeated. For each model a 40-window pilot, balanced on the label of the rule like that of gpt-6-luna, ran first.

Table 3. Experimental arms and run matrix. Windows, requests and cost come from the run ledger (Tables S1–S4 of the Supplementary Materials). Requests include the test-retest repeats and the 1,121 re-requests of 2026-11-03 (Section 3.7), and the pilot ran synchronously at standard prices. Batch costs use $0.05 and $0.25 per million input and output tokens. The three o4-mini backfills are not spend of this study. The arms with the prefix `oss_` and their pilot ran gpt-oss-20b on local GPUs (Section 3.5) and cost nothing; their requests include the 5% repeats and 14 retries at 8,000 tokens. The arms with the prefix `llama_` ran Llama 3.1 8B in the same way; none of its answers needed a retry.

Arm	Series	Symbol	Window ends	Windows	Requests	Cost (USD)
`r1_o4mini`	`jan2026_phase4a`	SPY	2020–2025	1,476	0 (backfill)	–
`r1_p3_o4mini`	`p3_2024_dated`	SPY	2024	223	0 (backfill)	–
`r1_p4_o4mini`	`p4_2020_dated`	SPY	2020	223	0 (backfill)	–
`bridge_jan`	`jan2026_phase4a`	SPY	2020–2025	1,476	2,952	0.82
`full_spy`	`new_full`	SPY	2008–2026	4,691	5,253	1.54
`ex0dte_spy`	`new_ex0dte`	SPY	2008–2026	4,691	5,237	1.50
`p34_obf_2024`	`p3_2024`	SPY	2024	223	234	0.06
`p34_dated_2024`	`p3_2024_dated`	SPY	2024	223	234	0.06
`p34_obf_2020`	`p4_2020`	SPY	2020	223	234	0.06
`p34_dated_2020`	`p4_2020_dated`	SPY	2020	223	234	0.06
`full_qqq`	`new_full`	QQQ	2011–2026	3,878	4,194	1.20
`full_iwm`	`new_full`	IWM	2008–2026	4,691	5,096	1.31
`full_qqq_scaled`	`new_full_scaled`	QQQ	2020–2026	1,671	1,785	0.54
`full_iwm_scaled`	`new_full_scaled`	IWM	2020–2026	1,672	1,785	0.50
`ex0dte_qqq`	`new_ex0dte`	QQQ	2020–2026	1,700	1,814	0.50
`ex0dte_iwm`	`new_ex0dte`	IWM	2020–2026	1,701	1,817	0.47
`intraday_spy`	`intraday_late`	SPY	2026-07 to 2026-10	51	54	0.02
`bridge_old`	`old_bs`	SPY	2008–2026	4,638	4,870	1.36
`pilot_bridge_jan`	`jan2026_phase4a`	SPY	2020–2025	40	40 (sync)	0.02
`oss_bridge_jan`	`jan2026_phase4a`	SPY	2020–2025	1,476	1,553	local
`oss_full_spy`	`new_full`	SPY	2008–2026	4,691	4,933	local
`oss_ex0dte_spy`	`new_ex0dte`	SPY	2008–2026	4,691	4,934	local
`oss_p34_obf_2024`	`p3_2024`	SPY	2024	223	234	local
`oss_p34_dated_2024`	`p3_2024_dated`	SPY	2024	223	236	local
`oss_p34_obf_2020`	`p4_2020`	SPY	2020	223	234	local
`oss_p34_dated_2020`	`p4_2020_dated`	SPY	2020	223	234	local
`pilot_oss_bridge_jan`	`jan2026_phase4a`	SPY	2020–2025	40	40	local
`llama_bridge_jan`	`jan2026_phase4a`	SPY	2020–2025	1,476	1,550	local
`llama_full_spy`	`new_full`	SPY	2008–2026	4,691	4,926	local
`llama_ex0dte_spy`	`new_ex0dte`	SPY	2008–2026	4,691	4,926	local
`llama_p34_obf_2024`	`p3_2024`	SPY	2024	223	234	local
`llama_p34_dated_2024`	`p3_2024_dated`	SPY	2024	223	234	local
`llama_p34_obf_2020`	`p4_2020`	SPY	2020	223	234	local
`llama_p34_dated_2020`	`p4_2020_dated`	SPY	2020	223	234	local
`pilot_llama_bridge_jan`	`jan2026_phase4a`	SPY	2020–2025	40	40	local
Total, gpt-6-luna					35,833	10.02
Total, gpt-oss-20b					12,398	local
Total, Llama 3.1 8B					12,378	local

The runs went in priority order. The first group was `bridge_jan`, `full_spy`, `ex0dte_spy` and the four `p34_*` arms; the second the unscaled and scaled arms of QQQ and IWM; the third the ex-0DTE arms of QQQ and IWM and the intraday arm; `bridge_old` was optional and ran last. Test-retest duplicates carry the suffix `#rep1`: `bridge_jan` is repeated in full (1,476 requests), and every other arm carries a duplicate sample of about 5% of its windows (1,516 requests in all). The first pass alone feeds every rate, and the duplicates feed the agreement of repeated runs. A request that ends with `finish_reason` = `length` is repeated once with 8,000 tokens; none was needed. All new requests cost $10.02 in total, of which $0.32 for the re-requests of 2026-11-03, against a hard stop of $12.00. Only runs paid in this study count: the three o4-mini backfills, worth $9.97 at o4-mini Batch prices, do not. The

scaling of QQQ and IWM was decided on 2026-10-28, and both variants are run side by side. It is a protocol change that is disclosed as such, not a re-tuning of the thresholds.

3.7. Storage and Reproducibility

Every series value, window, verdict of the rule, run and model answer is written to four tables, `gex_llm_regime_series`, `gex_llm_regime_windows`, `gex_llm_regime_runs` and `gex_llm_regime_detections`, with a summary view of detection rates per arm and year. The tables exist as parquet files under `data/backups/gex_llm/tables/`, and, after the documentation pull request in the database repository was approved, as standalone tables without foreign keys into the core tables in the `backups` schema of the project database; the schema was applied and the parquet copies were loaded on 2026-10-30. The raw batch input and output files, the batch object (for a local run, the record of the engine and its log) and a parquet copy of the detections are kept per run under `data/backups/gex_llm/<run_id>/`, with run identifiers of the form `<arm>_<model>_<mode>_<UTC timestamp>`. The three backfilled o4-mini runs are `r1_o4mini_phase4a_20260104`, `r1_p3_o4mini_20251120` and `r1_p4_o4mini_20251120`, and a test-retest repeat is stored under the arm `<arm>#rep1`. All tables and figures of this article are generated from the stored tables and from the stored Phase 3, 4 and 4A files; no figure falls back to synthetic data. Every window stores the SHA-256 of its input, every run row the hashes of its batch input and output files, and `report.yaml` the hashes of all result files and a fingerprint of the stored tables (beginning 11cdcadd82ec). The project keeps three copies of every result: the database tables, the parquet files and the raw batch files on the local disk. Nothing raw is in the repository or on its remote. The raw files and the full tables, which hold the model text, are ignored by git; the repository keeps the condensed tables (detections without text, windows without sequences, a series index, and the run provenance as one table of runs and one file of run records) and the result files of this article. Every detection rate and agreement figure of the article can be recomputed from the condensed tables; the daily values behind the magnitudes and the exposure shares are in the full tables. The database tables are the authoritative copy, and their parquet copies are deposited at Zenodo (Data Availability Statement). The run ledger is in Tables S1–S4 of the Supplementary Materials. After the database additions of 2026-11-03 the series and windows were rebuilt; every pre-existing day is unchanged. Each model answer whose 30-day input changed was requested again: 1,121 requests in eight Batch jobs on 2026-11-03. Of these, 1,049 replace earlier answers, which are recorded in `detection_metadata.supersedes` and no longer enter any rate, and the other 72 belong to windows that did not exist before. The earlier jobs stay in the run ledger, so the ledger counts 35,833 requests and the stored tables hold 34,784 current gpt-6-luna answers. The local runs of gpt-oss-20b add 12,398 requests and 12,384 current answers; the 14 answers that a retry at 8,000 tokens replaced are kept in the same way.

3.8. Statistics

Detection rates are reported with 95% Wilson intervals [33], and differences between rates with Newcombe–Wilson intervals [34]. Windows have stride one, so neighbouring windows share 29 of their 30 days and cannot be treated as independent. Every p-value is therefore accompanied by (i) a calendar-quarter block bootstrap [35], computed exactly over all equally likely resamples when they are few (up to ten quarters, or for two independent periods up to 200,000 distinct pairs of resamples) and from 2,000 draws (seed 261) otherwise and (ii) the same statistic on a non-overlapping 30-day subsample, the greedy chain of windows that share no day, starting at the first window of the period. Agreement between models is measured with Cohen's kappa [36], and paired detection rates across series or arms with the exact McNemar test [37] (on the subsample; the stride-one value is reported as anti-conservative); effect sizes between years use φ. The p-values of the AIAI and JRFM papers [1,3] were computed under independence of windows. The quarter-block bootstrap resamples the calendar quarters of the window end with replacement, and a paired contrast resamples the same quarters for both series. Its two-sided p-value is $2\min\{P(d \le 0), P(d \ge 0)\}$, bounded below by the probability

of one resample ($1/k^k$ for k quarters enumerated exactly, the product $1/(k_0^{k_0}k_1^{k_1})$ for two independent periods, and 1/2,001 for draws). With k quarters there are k^k equally likely resamples, so the interval of a single year (four quarters, 256 resamples) is coarse; enumerating the resamples makes it exact and independent of the seed. The period 2024 to 2025 gives 8 quarters. The non-overlapping subsample depends on its first window, so the 30 possible first windows give 30 chains, and the range of their rates is reported beside the rate of the chain that starts at the first window. For the rule on `new_full` (SPY, window ends in 2024 and 2025), the chain holds 3 of 17 windows (17.6%), and the rate ranges from 5.9% to 25.0% across the starts. Between periods, φ is signed (late minus early) and the p-value comes from Fisher's exact test. Each p-value in Section 4 names the method it uses.

4. Results

4.1. What the Published Inputs Were

Table 4 compares the inputs recovered from the Phase 4A batch file with each candidate series, and Figures 2 and 3 show one window and the 2008–2026 history. The batch file of 2026-01-04 stores the daily values that were sent to o4-mini. They give 1,476 windows with ends from 2020-02-13 to 2025-12-31, cut from 1,505 daily values. Compared with the complete series, the January calendar lacks 2023-07-12 and 2024-03-11. The 58 windows that span one of the two gaps are flagged and kept (Section 3.4).

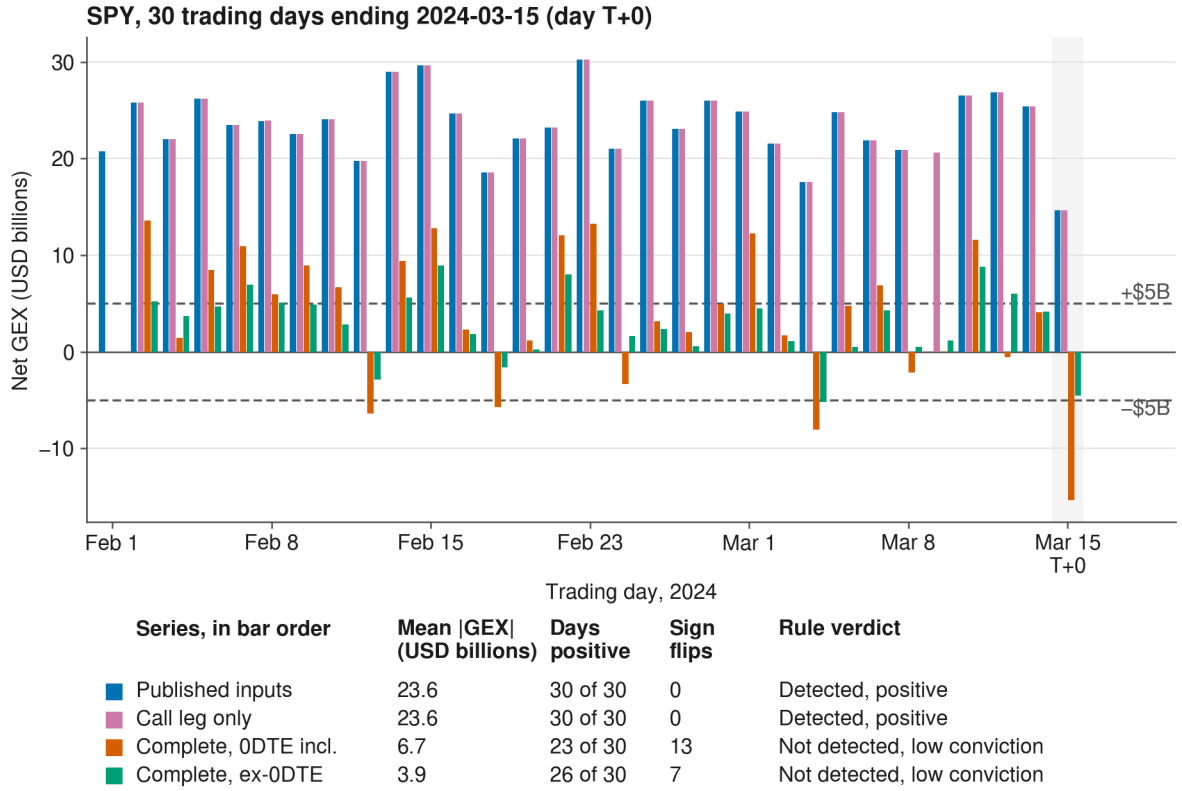
The published inputs follow two different series (Table 4). In 2020 to 2022 the share of daily values that equal `new_full` at the 0.01B precision of the prompt is 100.0%, 99.2% and 98.4%, and the call-only series equals the inputs on at most 1.2% of the days. In 2024 and 2025 the relation is reversed: none of the values equals `new_full` (0.0% in both years) and every value equals the call-only GEX (100.0% in both years). The year 2023 is the transition: 60.6% of its values equal `new_full` and 39.4% equal the call-only series, which together account for every day. From 2023-09-29 on, every value equals the call-only GEX; the day before, 2023-09-28, is the last on which a value differs from it. Before that date, 39 days (the first on 2021-09-17) already equal the call-only value and not the complete one. The published inputs of 2024 and 2025 are therefore the call leg of the options chain without the put leg. They differ in level and sign from the complete series (Table A1 gives the means): the mean |GEX| of the published inputs is \$20.19 billion in 2024 against \$6.48 billion on the complete series, and \$19.21 billion against \$7.13 billion in 2025, with the same daily sign on 62.2% and 54.0% of the days.

The pre-rebuild archive (`old_bs`) was computed from the same put-less table for the same dates, so it cannot serve as a clean earlier convention. Its Black–Scholes gamma differs from the stored gamma, and none of its values equals a published value at 0.01B (0.0% in every year). Its signs agree with the published inputs on 100.0% of the days in 2024 and 99.6% in 2025, and its mean |GEX| is \$16.19 billion and \$14.71 billion, which is the level of a call-only series. The archive also has holes: it lacks 53 days of the complete calendar (the 19 days of 2009 to 2018 and 2026-05-06 that were added on 2026-11-03, 2023-07-12, 2024-03-11, the 26 days from 2026-01-02 to 2026-02-09, and the 5 days from 2026-10-05 to 2026-10-09).

The inputs of the Phase 3 and 4 runs of November 2025 survive in the same way: 252 daily values for 2024 and 252 for 2020, 223 windows each, with ISO dates in the prompts. The 2024 values are a gross call-plus-put magnitude with a negative sign imposed. They are negative on 242 days, and the 10 positive values (\$10.61 billion to \$13.76 billion) are placeholder rows of the source database. The 2020 values are a magnitude only and are positive on every day. The mean absolute daily value is \$31.24 billion for 2024 and \$2.85 billion for 2020, so the two baselines differ in convention and in scale (Section 4.11).

Table 4. Agreement of the recovered published inputs with each candidate series, by year (share of daily values equal at the 0.01B precision of the prompt, in percent). The archive uses another gamma model, so exact equality is not expected for it.

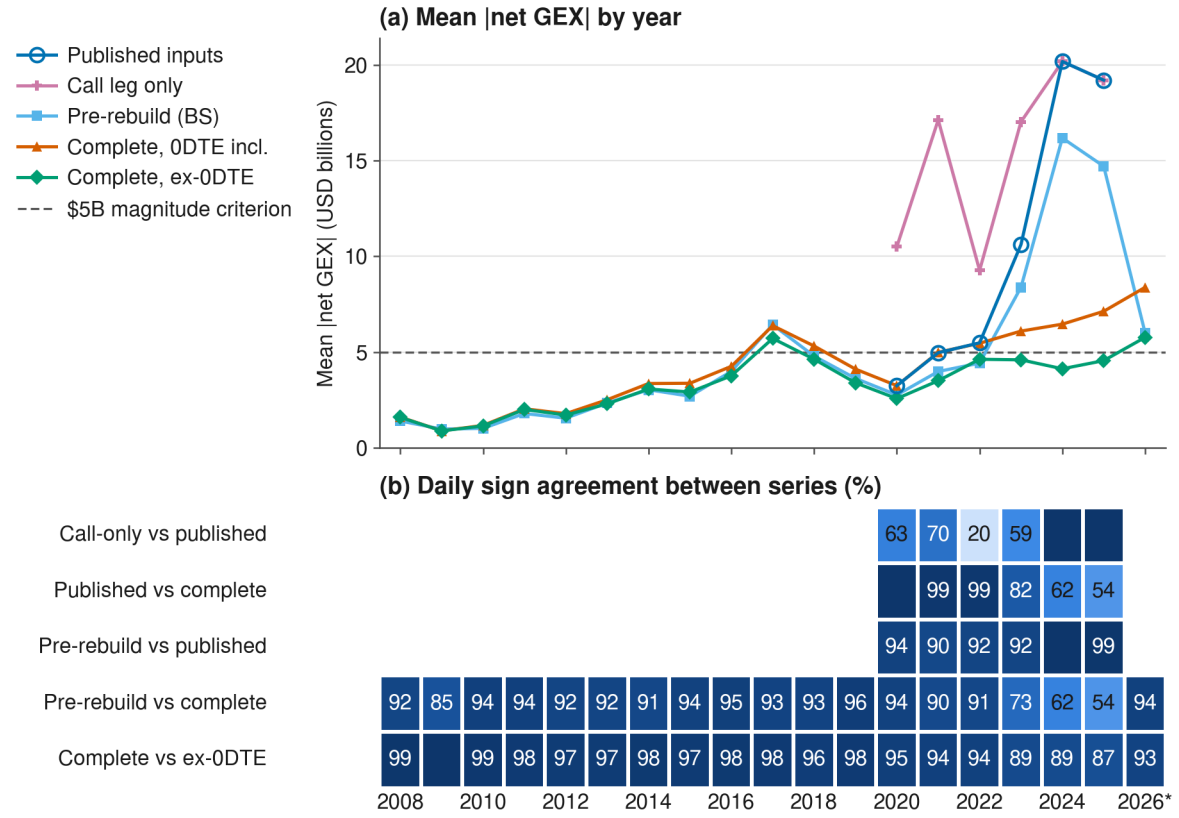
Year	Days	`new_full`	`call_only`	`old_bs`
2020	252	100.0	0.0	0.0
2021	252	99.2	0.8	0.0
2022	251	98.4	1.2	0.0
2023	249	60.6	39.4	0.0
2024	251	0.0	100.0	0.0
2025	250	0.0	100.0	0.0



Series, in bar order	Mean |GEX| (USD billions)	Days positive	Sign flips	Rule verdict
Published inputs	23.6	30 of 30	0	Detected, positive
Call leg only	23.6	30 of 30	0	Detected, positive
Complete, 0DTE incl.	6.7	23 of 30	13	Not detected, low conviction
Complete, ex-0DTE	3.9	26 of 30	7	Not detected, low conviction

Dashed lines: the $5B magnitude criterion of the classifier (mean |GEX| of the 30 days). Published inputs: no row for 2024-03-11, which the other series have, so its window starts on 2024-02-01.

Figure 2. One SPY 30-day window under four conventions (`jan2026_phase4a`, `call_only`, `new_ex0dte`, `new_full`), window ending 2024-03-15. Dashed lines mark the ±$5 billion magnitude threshold; the table under the bars gives the three criteria and the verdict for each convention. **Read this figure as:** the same 30 trading days receive different verdicts. The published inputs and the call-only series have 30 of 30 positive days, a mean |GEX| of $23.56 billion, 0 sign flips, and are detected. The complete series has 23 positive days, 13 flips and a mean of $6.66 billion, and the ex-0DTE series 26 positive days, 7 flips and $3.87 billion; both are rejected as low conviction. On the last day the net GEX is +$14.61 billion on the published inputs, −$15.35 billion on the complete series and −$4.51 billion without same-day expiries.



SPY, daily values. (a) Yearly mean over the days a series shares with its comparison partner; the published inputs are drawn hollow, on top. (b) Share of days on which both values are positive or both are not; dark cells without a number are 100%, blank cells have no days that year. * 2026: trading days up to 2026-10-09.

Figure 3. SPY, 2008–2026. (a) Yearly mean of |GEX| under each series, with the $5 billion magnitude criterion for scale. (b) Daily sign agreement between pairs of series, by year. **Read this figure as:** until 2022 the published inputs and the complete series are close; from 2023 the published inputs rise to $20.19 billion in 2024 while the complete series stays between $6.05 billion and $7.13 billion in 2023 to 2025. Their daily sign agreement falls from 98.8% in 2022 to 81.5%, 62.2% and 54.0%. The call-only series matches the published inputs in 2024 and 2025 only.

4.2. Classifier Replay by Year

Table 5 and Figure 4 give the detection rate of the deterministic rule by year for each series next to the published o4-mini rates (per-year values for all series are in Table A2), and Figure 5 shows how far the replay depends on the thresholds. On the recovered inputs the rule reproduces the published labels in 1,412 of 1,412 parsed windows (100.0%; Cohen's kappa 1.000), so the published labels follow from the published inputs and the published criteria. On the published inputs the rule detects every window ending in 2024 and 2025, and so do the published o4-mini answers (Table 5; Wilson 95% lower bound 98.4% in 2024 and 98.5% in 2025). The call-only series and the archive give the same saturation. On the complete series the rule detects 10.3% in 2024 and 13.6% in 2025 (Table 5; quarter-block bootstrap 3.2–17.3% and 2.3–25.4%, 4 blocks each).

Pooled over 2024 and 2025 the detection rate is 100.0% on the published inputs (501 of 501 windows) and 12.0% on the complete series (60 of 502; Wilson 9.4–15.1%, quarter-block bootstrap 5.1–19.6%; non-overlapping subsample 3 of 17, and between 5.9% and 25.0% over the 30 possible starting windows). The paired difference is 88.0 percentage points (quarter-block bootstrap interval 80.4–94.9, $p = 6.0 \times 10^{-8}$, the smallest value the exact enumeration of the 8^8 resamples can give, because no resample reaches a difference of zero; exact McNemar test on the non-overlapping subsample $p = 1.2 \times 10^{-4}$).

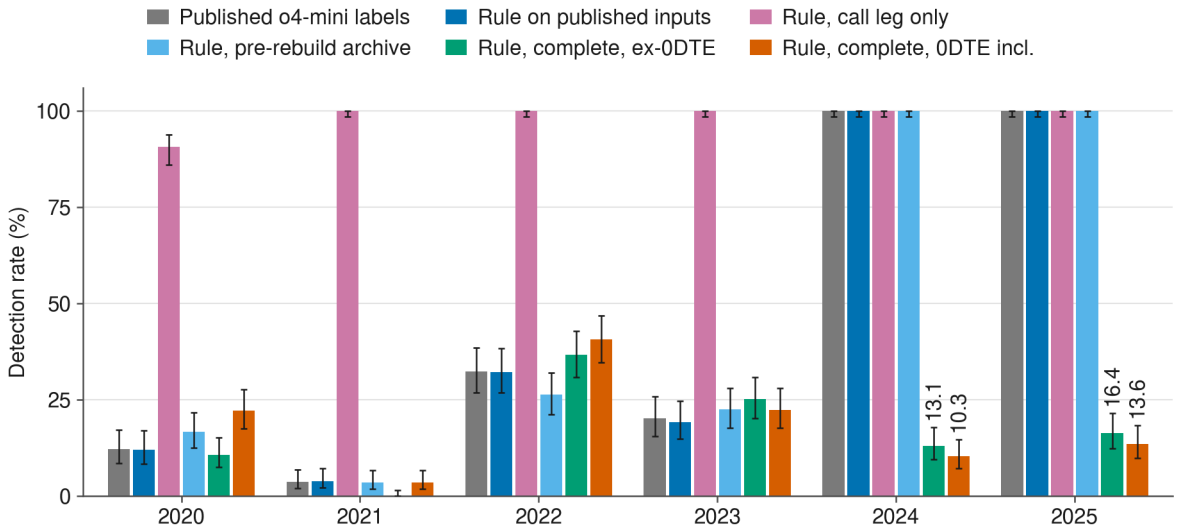
The detections also differ in sign. On the published inputs all 501 detected windows of 2024 and 2025 are persistent positive-GEX windows. The call leg alone is positive by construction (Equation (1)): each of its 1,672 windows has 30 positive days and no sign change, so the magnitude criterion alone decides, and it fails only in the 21 windows that end from 2020-04-02 to 2020-05-01. On the complete series the 60 detections of 2024 and 2025 are 30 positive windows (26 ending in 2024 and 4 in 2025) and 30 negative windows, all ending between 2025-03-18 and 2025-04-29. The detections of 2022 (102 windows) and of 2026 (22 windows; Section 4.8) are all negative. The published inputs therefore implied a positive regime in every window of 2024 and 2025, which the complete series does not show.

The period contrast changes sign. On the published inputs the rule detects 17.0% of the windows of 2020 to 2023 and 100.0% of those of 2024 and 2025, a shift of 83.0 percentage points ($\varphi = 0.789$; bootstrap $p = 5.0 \times 10^{-4}$; Fisher exact test on the non-overlapping subsample $p = 2.7 \times 10^{-9}$). On the complete series the rates are 22.2% and 12.0%, a shift of -10.2 percentage points ($\varphi = -0.124$; bootstrap $p = 0.108$; Fisher exact test on the subsample $p = 0.728$): the complete series shows no increase after 2023 that the data distinguish from zero. In 2020 to 2023 the two series are of the same order (Table 5).

Across the 45 threshold sets the rate on the published inputs stays at 100.0%, whereas the rate on the complete series runs from 4.0% to 29.3%. The gap between the two is at least 70.7 and at most 96.0 percentage points in every configuration, so the difference between the published and the complete inputs does not depend on the exact thresholds.

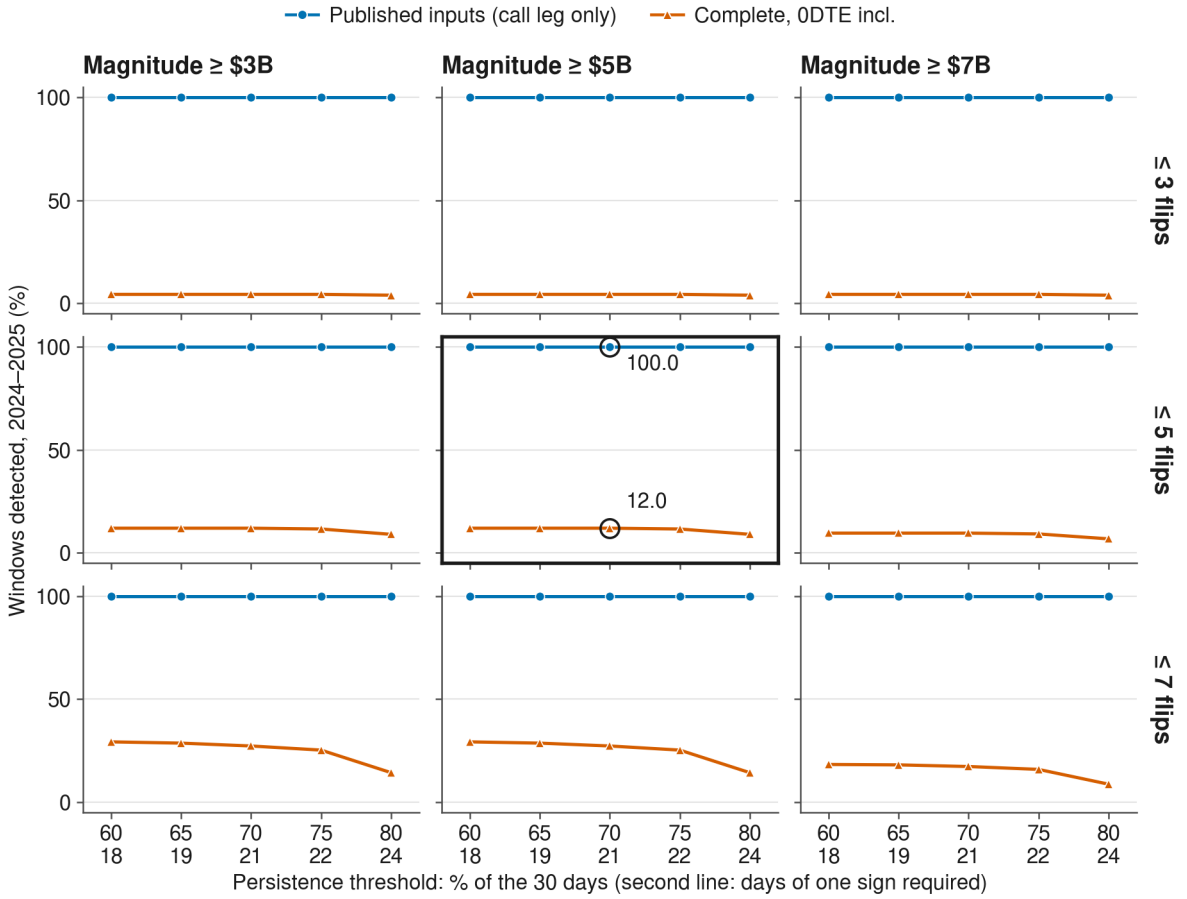
Table 5. Detection rate of the deterministic classifier by year, SPY, in percent, next to the published o4-mini rate. Each cell gives the rate and, in parentheses, the windows detected out of the windows classified. The denominator of the `new_full` cell is the number of windows ending in the year; the other columns can hold fewer. The published rate covers the 1,412 answers of the published result.

Year	Published	`jan2026_phase4a`	`call_only`	`old_bs`	`new_ex0dte`	`new_full`
2020	12.2 (26/213)	12.1 (27/223)	90.6 (202/223)	16.7 (42/252)	10.7 (27/252)	22.2 (56/252)
2021	3.7 (9/241)	4.0 (10/252)	100.0 (252/252)	3.6 (9/252)	0.0 (0/252)	3.6 (9/252)
2022	32.4 (79/244)	32.3 (81/251)	100.0 (251/251)	26.3 (66/251)	36.7 (92/251)	40.6 (102/251)
2023	20.2 (46/228)	19.3 (48/249)	100.0 (250/250)	22.5 (56/249)	25.2 (63/250)	22.4 (56/250)
2024	100.0 (241/241)	100.0 (251/251)	100.0 (252/252)	100.0 (251/251)	13.1 (33/252)	10.3 (26/252)
2025	100.0 (245/245)	100.0 (250/250)	100.0 (250/250)	100.0 (250/250)	16.4 (41/250)	13.6 (34/250)



SPY, 30-day windows by year of the window end. Error bars: Wilson 95% CI (overlapping windows treated as independent, so optimistic); a year holds only 8–9 non-overlapping windows. First bar: o4-mini output of January 2026 (parsed answers for 1,412 of 1,476 windows; the 64 answers re-parsed in 2026 are not drawn); other bars: the rule applied to all windows of the series in the legend.

Figure 4. Detection rate by year, SPY, 2020–2025: the published o4-mini labels and the deterministic rule replayed on the published inputs (`jan2026_phase4a`), the call leg only (`call_only`), the pre-rebuild archive (`old_bs`) and the complete series without and with same-day expiries (`new_ex0dte`, `new_full`), with 95% Wilson intervals. **Read this figure as:** the published labels and the rule on the published inputs agree window by window. In 2024 and 2025 the bars of the published inputs, the call-only series and the archive stand at 100%, and the two bars of the complete series stand at 13.1% and 10.3% in 2024 and 16.4% and 13.6% in 2025. The Wilson intervals are optimistic because the overlapping windows of a year are treated as independent.



SPY windows ending in 2024 or 2025 (501 on the published inputs, 502 on the complete series), each re-classified from its stored 30-day sequence. Heavy frame and rings: the published thresholds (70%, $5B, 5 flips). Across the 45 threshold sets the published-input rate exceeds the complete rate by 70.7 to 96.0 percentage points.

Figure 5. Sensitivity of the 2024–2025 detection rate to the persistence, magnitude and flip thresholds (45 threshold sets: persistence 60, 65, 70, 75 and 80%, magnitude $3, $5 and $7 billion, flip limit 3, 5 and 7; one panel per magnitude and flip limit, persistence on the horizontal axis) for the published inputs (`jan2026_phase4a`) and the complete series (`new_full`); the panel with the heavy frame holds the published thresholds. **Read this figure as:** the curve of the published inputs is flat at 100.0% in all nine panels, and the curve of the complete series stays low and moves with the flip limit: it is highest, up to 29.3%, with a limit of seven flips and lowest, 4.0%, with three. At the published thresholds the complete series gives 12.0%.

4.3. Model Agreement on the Published Inputs

Table 6 and Figure 6 compare gpt-6-luna with the published o4-mini answers on identical inputs. The agreement gate of Section 3.5 (Cohen's kappa of at least 0.6) is met with a wide margin. On the 1,412 windows with a published answer the two models give the same verdict in 1,407 windows (99.6%; quarter-block bootstrap interval 99.4–99.9%), and kappa is 0.993 (bootstrap interval 0.986–0.998). The confusion matrix has 646 windows detected by both, 761 by neither, 0 only by o4-mini and 5 only by gpt-6-luna. In each of the 5 windows of disagreement the deterministic rule agrees with o4-mini. The windows lie in 2021 to 2023 (3 in 2021, 1 in 2022, 1 in 2023); the yearly kappa is lowest in 2021 (0.851) and between 0.986 and 1.000 in 2020, 2022 and 2023, and it is undefined in 2024 and 2025, where both models label every window as detected. The detection rates are 45.8% for o4-mini and 46.1% for gpt-6-luna on these windows. The paired McNemar test on all windows at stride one gives $p = 0.063$ (anti-conservative, because the windows overlap) and the quarter-block bootstrap $p = 0.010$ (Table 6 gives both).

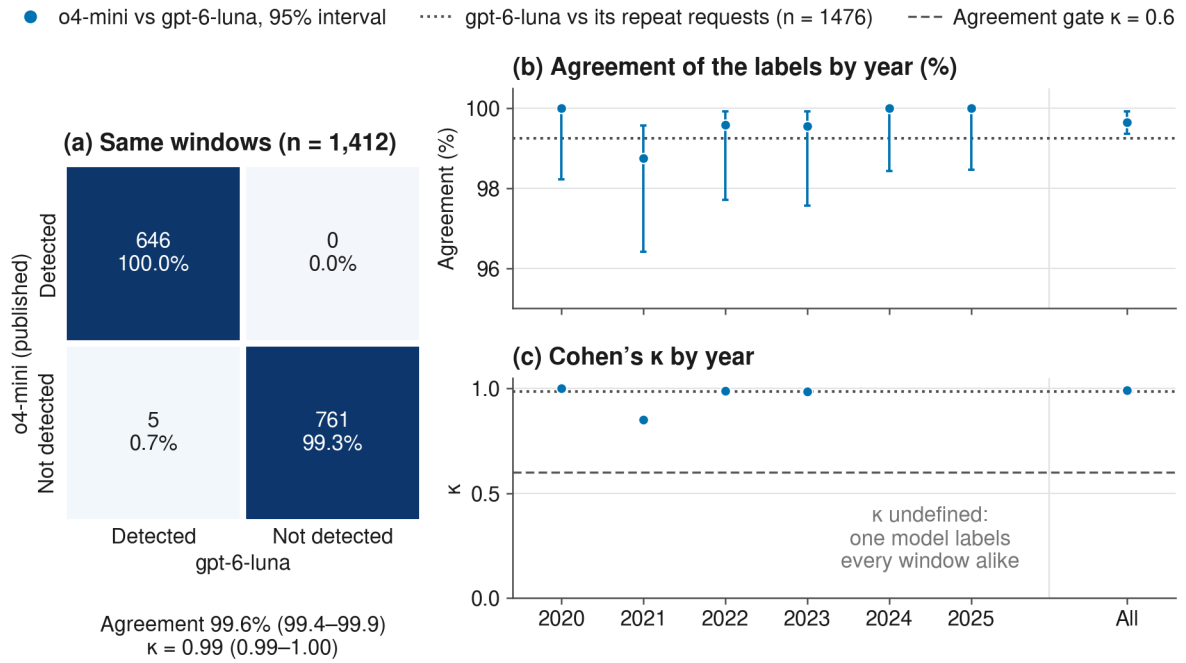
The 64 answers of the published run that were lost to a parsing error in January 2026 and re-parsed from the stored raw output are held out of every headline number. With them the comparison covers 1,476 windows: agreement 99.6%, kappa 0.992 (bootstrap 0.984–0.997), and 6 windows of disagreement, all with the rule on the side of o4-mini. The conclusions do not depend on whether they are included.

On the exact published inputs the mean magnitude that o4-mini reports equals the mean computed from the window: the median ratio of reported to computed is 1.000, and 99.9% of the answers lie within 5% of the computed value. For gpt-6-luna the median ratio is 1.000 (99.7% within 5%). The pre-rebuild archive cannot serve for this check: its daily values are 0.818 times the published inputs at the median (ratio of the archive to the published inputs), so a comparison with the archive would suggest a magnitude error that the exact inputs rule out.

The repeat pass runs gpt-6-luna a second time on all 1,476 windows of the published inputs. The two passes agree on 99.3% of the windows (kappa 0.985, bootstrap 0.969–0.996) and differ in 11 windows, which is more than the 5 windows in which the first pass differs from o4-mini. The disagreement between the two models is thus no larger than the disagreement of the model with itself. The repeats of the other arms agree on 98.8% to 100.0% of their windows (Table S1 of the Supplementary Materials lists the runs; the repeat of `full_spy` agrees on 99.1% of 235 windows).

Table 6. Bridge to the published result: gpt-6-luna versus o4-mini on the exact Phase 4A inputs. The first block holds the 1,412 answers of the published result, the second adds the 64 re-parsed answers, and the third compares gpt-6-luna with its own repeat pass. Intervals are quarter-block bootstrap intervals; the McNemar test is exact on all windows at stride one and therefore optimistic.

Statistic	o4-mini	gpt-6-luna	Both
Windows with a parsed answer	1,412	1,412	1,412
Detection rate (%)	45.8	46.1	–
Agreement with the classifier (%)	100.0	99.6	–
Agreement between models (%) [95% block-bootstrap interval]	–	–	99.6 [99.4, 99.9]
Cohen's kappa [95% block-bootstrap interval]	–	–	0.993 [0.986, 0.998]
McNemar exact p (stride one, all windows)	–	–	0.063
Block-bootstrap p (calendar quarters)	–	–	0.010
Median reported / computed mean |GEX|	1.000	1.000	–
With the 64 re-parsed answers			
Windows with a parsed answer	1,476	1,476	1,476
Detection rate (%)	45.2	45.5	–
Agreement with the classifier (%)	100.0	99.6	–
Agreement between models (%) [95% block-bootstrap interval]	–	–	99.6 [99.3, 99.9]
Cohen's kappa [95% block-bootstrap interval]	–	–	0.992 [0.984, 0.997]
McNemar exact p (stride one, all windows)	–	–	0.219
Block-bootstrap p (calendar quarters)	–	–	0.121
Test-retest of gpt-6-luna (first pass against repeat pass)			
Windows with a parsed answer	–	1,476	–
Agreement between passes (%) [95% block-bootstrap interval]	–	–	99.3 [98.6, 99.8]
Cohen's kappa [95% block-bootstrap interval]	–	–	0.985 [0.969, 0.996]



SPY windows of the recovered Phase 4A inputs, window ends 2020–2025. Whiskers on years: Wilson 95% intervals, which treat overlapping windows as independent; All: quarter-block bootstrap 95% interval. In (a) the percentage is the share of the row.

Figure 6. Bridge to the published result on the Phase 4A windows. (a) Confusion matrix of gpt-6-luna against the o4-mini labels, with the overall agreement and Cohen's kappa. (b), (c) Agreement and kappa by year, with the test-retest of gpt-6-luna and the agreement gate as dashed references. **Read this figure as:** nearly all windows lie on the diagonal (646 and 761), and the five off-diagonal windows are all detections by gpt-6-luna alone. Yearly agreement is 98.8% or higher and the yearly kappa is far above the gate of 0.6, at 0.851 in its lowest year. The test-retest agreement of gpt-6-luna with itself (99.3%) is a lower reference than the agreement with o4-mini.

4.4. gpt-6-luna on the Complete Series

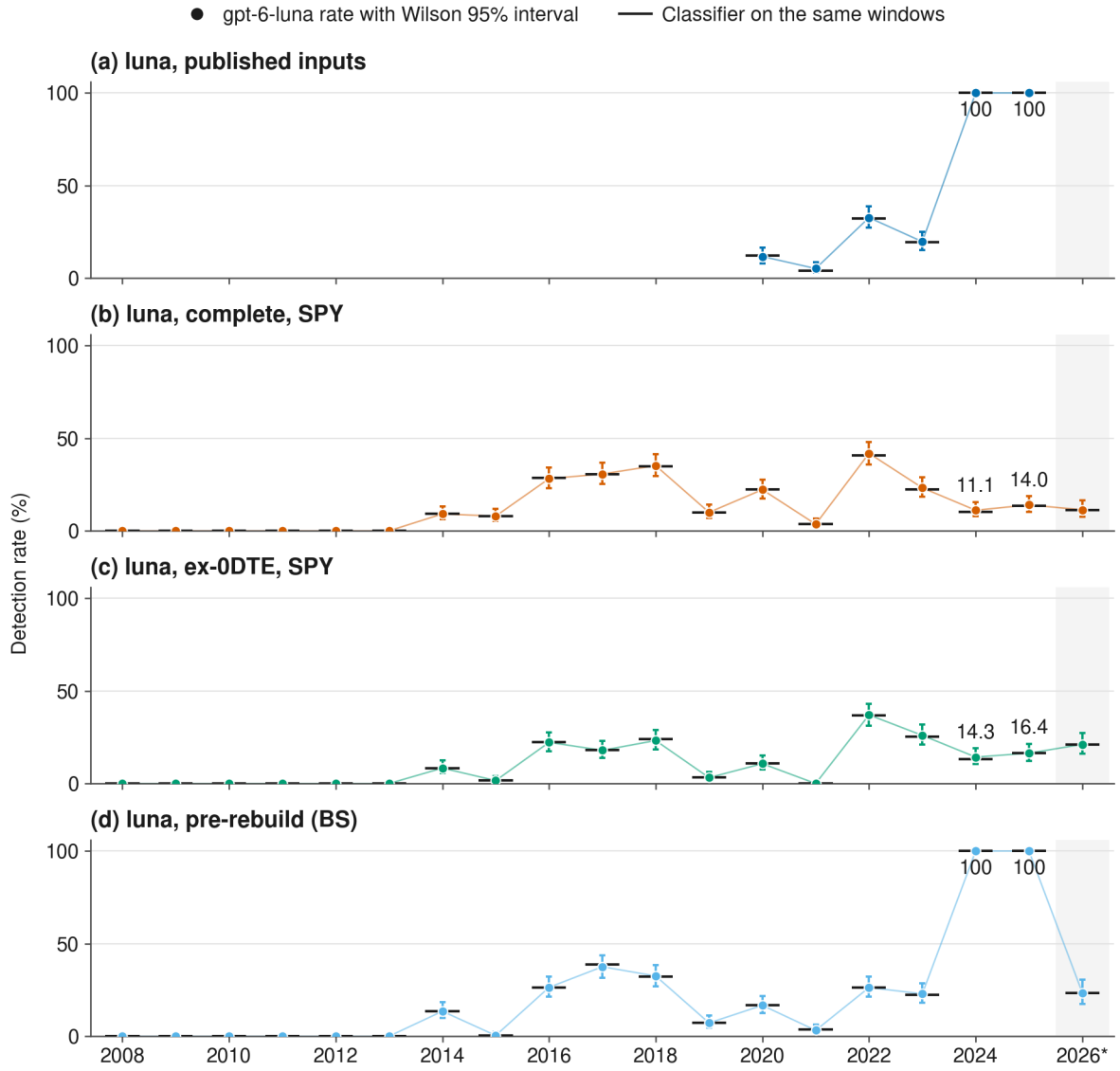
Figure 7 shows the detection rate of gpt-6-luna by year on each SPY series, from the published inputs to the complete series, next to the rule, and Table 7 gives the yearly rates of the two complete series with their intervals; Table A3 gives the QQQ and IWM arms. On the complete series gpt-6-luna labels 617 of 4,691 windows as persistent (13.2%; 609 of them by the rule, 13.0%) and, without same-day expiries, 495 of 4,691 (10.6%; rule 491, 10.5%). In 2024 and 2025 it detects 11.1% and 14.0% of the windows of the complete series and 14.3% and 16.4% of those without same-day expiries (Wilson intervals in Table 7). Pooled over the two years the rate is 12.5% (63 of 502; Wilson 9.9–15.7%, quarter-block bootstrap 5.9–19.9%, non-overlapping subsample 3 of 17). The same model on the published inputs detects 100.0% of the windows of these two years (501 of 501), and on the pre-rebuild archive 100.0% (501 of 501). The paired difference between the published inputs and the complete series is 87.4 percentage points (quarter-block bootstrap interval 80.1–94.1, $p = 6.0 \times 10^{-8}$; exact McNemar test on the non-overlapping subsample $p = 1.2 \times 10^{-4}$), and that between the archive and the complete series is 87.4 percentage points ($p = 6.0 \times 10^{-8}$ and $p = 1.2 \times 10^{-4}$). In 2020 to 2023 the model detects 22.7% of the windows of the complete series. In 2026 it detects 11.3% of the windows of the complete series and 21.1% of those without same-day expiries (Section 4.8).

The model follows the rule on the complete series. The verdicts agree in 99.7% of the 4,691 windows of `full_spy` (kappa 0.985) and in 99.7% of those of `ex0dte_spy` (kappa 0.984). Where the model and the rule both detect a regime, they give it the same sign in every window (605 of 605 on `full_spy` and 486 of 486 on `ex0dte_spy`). The two arms disagree with the rule in 16 and 14 windows; the causes are analysed in Section 4.10. All 34,784 stored gpt-6-luna answers were parsed at the first

attempt: every answer ended with finish reason `stop`, and none needed a fallback parse or the retry at the larger token limit.

Table 7. gpt-6-luna on the complete SPY series, by year of the window end: windows, detection rate in percent with its 95% Wilson interval (overlapping windows treated as independent, hence optimistic) for `full_spy` and `ex0dte_spy`, and the agreement of the model verdict with the rule on the same windows (percent). * Windows ending in 2026 up to 2026-10-09.

		`full_spy`		`ex0dte_spy`		Agreement with rule	
Year	Windows	Rate	Wilson 95%	Rate	Wilson 95%	full	ex-0DTE
2008	224	0.0	0.0–1.7	0.0	0.0–1.7	100.0	100.0
2009	252	0.0	0.0–1.5	0.0	0.0–1.5	100.0	100.0
2010	252	0.0	0.0–1.5	0.0	0.0–1.5	100.0	100.0
2011	252	0.0	0.0–1.5	0.0	0.0–1.5	100.0	100.0
2012	250	0.0	0.0–1.5	0.0	0.0–1.5	100.0	100.0
2013	252	0.0	0.0–1.5	0.0	0.0–1.5	100.0	100.0
2014	252	9.1	6.2–13.3	8.3	5.5–12.4	100.0	100.0
2015	252	7.9	5.2–11.9	1.6	0.6–4.0	100.0	100.0
2016	252	28.2	23.0–34.0	22.2	17.5–27.8	99.6	100.0
2017	251	30.7	25.3–36.6	17.9	13.7–23.1	99.2	99.2
2018	250	35.2	29.5–41.3	23.2	18.4–28.8	99.6	99.2
2019	251	10.0	6.8–14.3	3.2	1.6–6.2	100.0	100.0
2020	252	22.2	17.5–27.8	10.7	7.5–15.1	100.0	100.0
2021	252	3.6	1.9–6.6	0.0	0.0–1.5	99.2	100.0
2022	251	41.8	35.9–48.0	37.1	31.3–43.2	98.8	99.6
2023	250	23.2	18.4–28.8	26.0	21.0–31.8	99.2	98.4
2024	252	11.1	7.8–15.6	14.3	10.5–19.1	98.4	98.8
2025	250	14.0	10.2–18.8	16.4	12.3–21.5	99.6	99.2
2026*	194	11.3	7.6–16.6	21.1	16.0–27.4	100.0	100.0



SPY, 30-day windows ending in the year. Whiskers are Wilson intervals that treat the overlapping windows of a year as independent, so they are too narrow: a year holds about 9 non-overlapping 30-day windows. Grey band: 2026, the post-publication period (windows ending up to 2026-10-09).

Figure 7. gpt-6-luna detection rate by year, SPY: one panel per arm, from the published inputs (`bridge_jan`) to the complete series with and without same-day expiries (`full_spy`, `ex0dte_spy`) and the pre-rebuild archive (`bridge_old`). Dots are the model with 95% Wilson intervals, dashes the classifier on the windows the model answered, and the grey band is 2026. **Read this figure as:** on the published inputs (a) and on the archive (d) the rate jumps to 100% in 2024 and 2025. On the two complete series (b, c) it stays between 3.6% and 41.8% (`full_spy`) and between 0.0% and 37.1% (`ex0dte_spy`) from 2020 to 2026, with the highest value in 2022. In every panel the dots lie close to the dashes of the classifier.

4.5. 0DTE Ablation

Table 8 separates the contribution of same-day expiries by comparing `new_full` with `new_ex0dte`. The share of same-day expiries in the absolute net GEX of SPY grows from 0.9% in 2008 and 11.5% in 2013 to 22.1% in 2019, 25.7% in 2020 and 45.9% in 2024, and it stands at 43.9% in 2025 and 40.8% in 2026 (the share is the sum of the absolute same-day-expiry contribution over the sum of the absolute net GEX of `new_full` over the days of the year). For QQQ it is 45.5% in 2025 and for IWM 23.0%. The

sign of the daily net GEX changes when same-day expiries are removed on 11.1% of the days of 2024 and 13.2% of the days of 2025 (4.4% of all days).

The effect on the verdict is small and has no constant direction. In 2024 the rule detects 10.3% of the windows of the complete series and 13.1% of those without same-day expiries; 9 windows are detected only on the complete series and 16 only without same-day expiries (difference -2.8 percentage points; quarter-block bootstrap interval $[-10.7, 5.5]$, $p = 0.516$; exact McNemar test on the non-overlapping subsample $p = 1.000$). The counts for 2025 are 4 and 11 (rates in Table 8). Pooled over 2024 and 2025 the rates are 12.0% and 14.7% for the rule (difference -2.8 percentage points, interval $[-8.7, 2.8]$, $p = 0.361$; McNemar on the subsample of 17 windows $p = 1.000$) and 12.5% and 15.3% for gpt-6-luna (-2.8 percentage points, $p = 0.359$; subsample $p = 1.000$). Removing same-day expiries therefore does not restore the high rates of the published inputs: the rate is 14.7% without them and 12.0% with them, against 100.0% on the published inputs. In 2020 to 2023 the complete series detects more than the ex-0DTE series (22.2% against 18.1%, 4.1 percentage points; $p = 0.137$ by bootstrap, $p = 1.000$ by McNemar on the subsample), and in 2026 it detects less (11.3% against 21.1%; 0 windows against 19 in the paired comparison; bootstrap $p = 0.633$, McNemar on the subsample $p = 1.000$). For QQQ and IWM the rule detects no window under either series, so the ablation has no effect on their verdicts.

The window of Figure 2 gives an example. The net GEX of SPY on 2024-03-15 is $+\$14.61$ billion on the published inputs and $+\$14.61$ billion as call-only GEX, $-\$4.51$ billion without same-day expiries and $-\$15.35$ billion on the complete series. The archive value is $+\$12.62$ billion and is put-less too, so the step from the archive to the complete series cannot be read as an effect of the gamma model alone: it combines the restored put leg with the change of the gamma model.

Table 8. 0DTE ablation, SPY, by year: share of the absolute net GEX contributed by same-day expiries (percent), and detection rates (percent) of the classifier and of gpt-6-luna on `new_full` and `new_ex0dte`. The McNemar p-values are exact tests on the non-overlapping subsample (Section 3.8), reported for 2024, 2025 and 2026 and for the pooled periods 2008–2019, 2020–2023 and 2024–2025; the other years have no test (–).

	0DTE share	Classifier		gpt-6-luna		McNemar p	
Year	of |GEX|	full	ex-0DTE	full	ex-0DTE	classifier	gpt-6-luna
2020	25.7	22.2	10.7	22.2	10.7	–	–
2021	36.0	3.6	0.0	3.6	0.0	–	–
2022	23.4	40.6	36.7	41.8	37.1	–	–
2023	35.8	22.4	25.2	23.2	26.0	–	–
2024	45.9	10.3	13.1	11.1	14.3	1.000	1.000
2025	43.9	13.6	16.4	14.0	16.4	1.000	1.000
2026	40.8	11.3	21.1	11.3	21.1	1.000	1.000
2008–2019 (pooled)	–	10.2	6.5	10.2	6.4	0.375	0.375
2020–2023 (pooled)	–	22.2	18.1	22.7	18.4	1.000	0.688
2024–2025 (pooled)	–	12.0	14.7	12.5	15.3	1.000	1.000

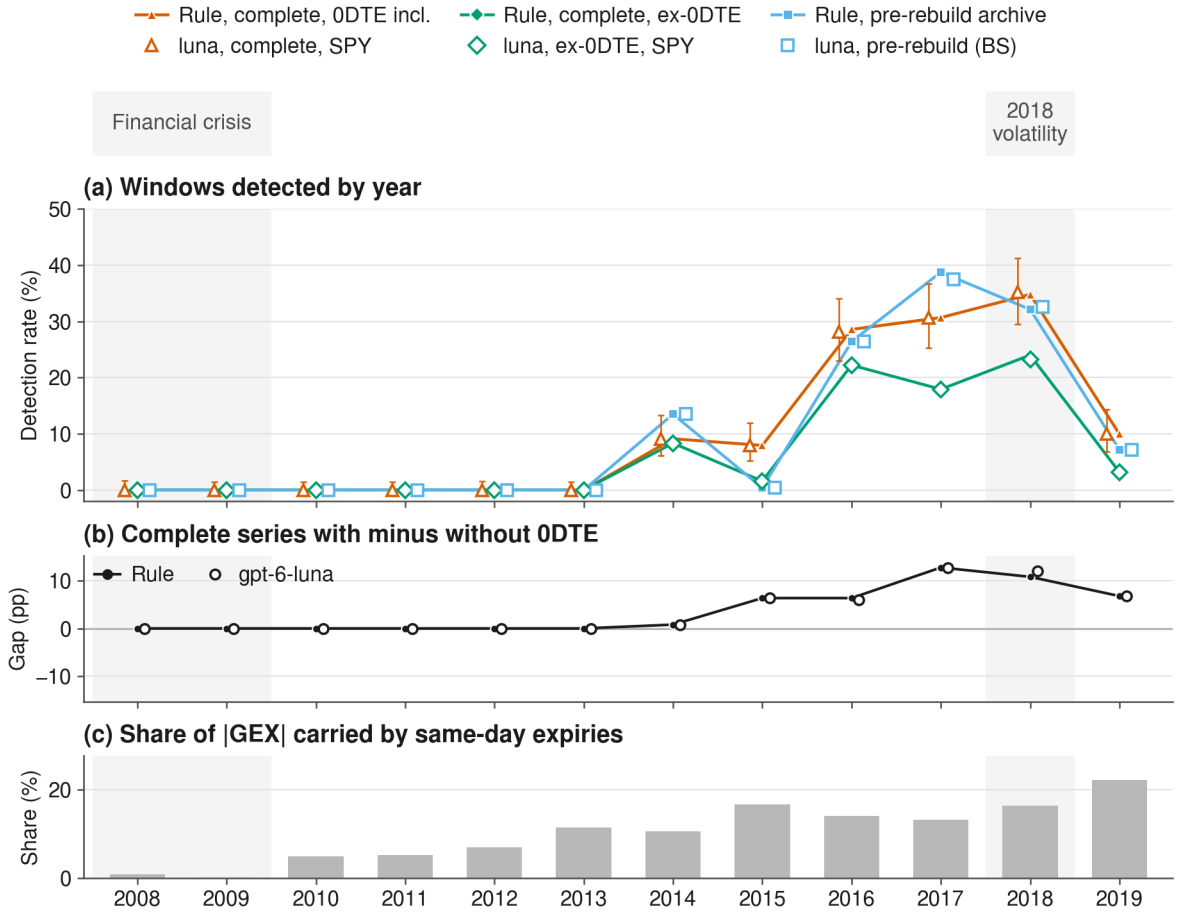
4.6. Deep Baseline, 2008–2019

Figure 8 extends the analysis to 2008–2019, before the growth of weekly and daily expirations. Before 2014 no window of any of the three series is detected, by the rule or by gpt-6-luna; this includes the 476 windows that end in the financial crisis of 2008 and 2009. No window of these years reaches the magnitude criterion: the largest 30-day mean |GEX| is $3.91 billion on `new_full`, $3.65 billion on `new_ex0dte` and $4.03 billion on the archive. The zero is therefore a floor effect of the fixed dollar threshold, as for QQQ and IWM (Section 4.7), and not an absence of persistent signs: 772 of the 1,482 windows of `new_full` that end in 2008 to 2013 (52.1%) meet the persistence and stability criteria, 703 of them with a negative sign, and fail on magnitude alone. Detections start in 2014, the first year in which a window mean reaches $5 billion (9.1% on `new_full`), and peak in 2018 on both complete series,

at 34.8% on `new_full` (87 of 250 windows) and 24.0% on `new_ex0dte`, the year of the 2018 volatility episode; the archive peaks in 2017 (38.8%). Pooled over 2008 to 2019 the rule detects 10.2% of the windows of `new_full` (304 of 2,990; Wilson 9.1–11.3%, quarter-block bootstrap 4.9–16.7%), 6.5% of those of `new_ex0dte` and 9.8% of those of the archive. The model gives 10.2%, 6.4% and 9.8% on the same three series.

The fixed threshold is not constant in real terms. GEX grows with the square of the price (Equation (1)), and the mean price of SPY rose from \$95 in 2009 to \$724 in 2026, so that factor alone grew 58-fold, besides the growth of open interest. Without the magnitude criterion, the share of windows of `new_full` with a persistent sign falls over the whole period: for SPY from 52.1% in 2008 to 2013 to 41.9% in 2014 to 2019, 28.8% in 2020 to 2023 and 12.0% in 2024 and 2025 (11.3% in 2026), and in the same direction for QQQ (51.0% in 2011 to 2013, 13.4% in 2024 and 2025) and IWM (80.0% and 35.7%). From 2023 on the magnitude criterion does not bind for SPY, so the share equals the detection rate and the comparison of Section 4.2 is unaffected.

The gap between `new_full` and `new_ex0dte` is a consistency check of the ablation arm. It is zero from 2008 to 2013, where neither series reaches the magnitude criterion, so these years cannot test the ablation, and it is positive in every year from 2014 to 2019: 0.8 percentage points in 2014, 6.3 in 2015, up to 12.7 in 2017 and 6.8 in 2019, while the share rises to 22.1%. The gap is therefore not proportional to the share. The model reproduces it (up to 12.7 percentage points). Over 2008–2019 the complete series detects 3.7 percentage points more than the ex-0DTE series (120 windows detected only with same-day expiries against 10 only without; quarter-block bootstrap $p = 5.0 \times 10^{-4}$; McNemar on the non-overlapping subsample of 100 windows $p = 0.375$). Only the bootstrap separates the two series in the decade before 0DTE; the exact test on the non-overlapping subsample, with 4 against 1 discordant windows, cannot ($p = 0.375$), whereas the sign of the difference is reversed, without a significant difference, in 2024 and 2025 (Section 4.5).



SPY, 30-day windows ending in the year. Lines: deterministic rule; open markers: gpt-6-luna on the same windows (shifted sideways for legibility). Whiskers on the first series: Wilson 95% interval, which treats the overlapping windows of a year as independent and is too narrow: a year holds about 9 non-overlapping 30-day windows. (b) Detection rate with 0DTE minus without, on the windows both series have. Shaded: 2008–2009 financial crisis and 2018 volatility episode.

Figure 8. Pre-0DTE baseline, SPY, 2008–2019, with the 2008–2009 financial crisis and the 2018 volatility episode marked. (a) Detection rate by year of the classifier (lines) and gpt-6-luna (markers). (b) Gap between `new_full` and `new_ex0dte`. (c) Share of |GEX| carried by same-day expiries. **Read this figure as:** all three series stay at zero until 2013, when no 30-day mean reaches the $5 billion criterion, and diverge as the same-day share in panel (c) rises, with the largest gap of 12.7 percentage points in 2017. The markers of gpt-6-luna lie close to the lines of the rule in every year.

4.7. Cross-Asset Replication and the Floor Effect

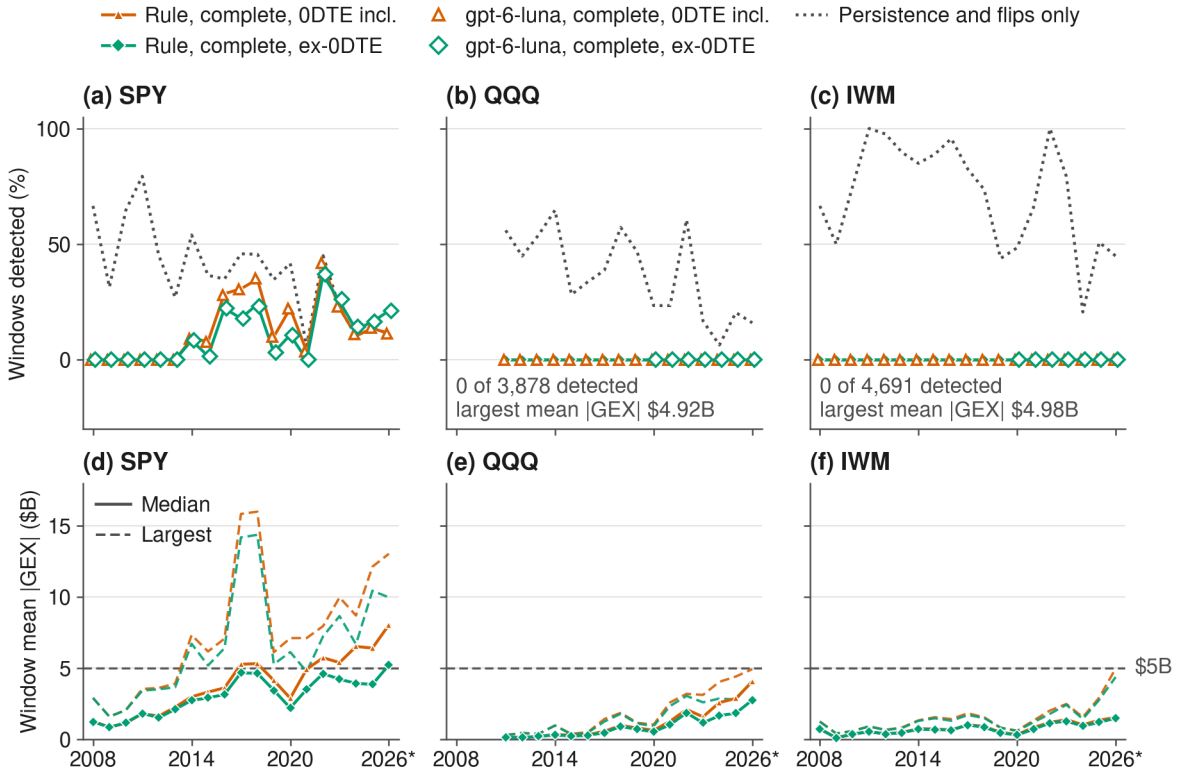
Table 9 and Figure 9 repeat the analysis for QQQ and IWM. The fixed dollar thresholds of Section 3.1 can produce a floor effect for smaller underlyings, which we report; the scaled series of Section 3.6 show whether the model's verdict still follows the criteria when the floor is lifted. The floor effect is complete. Under the fixed thresholds the rule detects no window of QQQ (0 of 3,878, windows ending 2011 to 2026-10-09) and none of IWM (0 of 4,691, 2008 to 2026-10-09), in any year and on either series. The largest 30-day mean |GEX| is $4.92 billion for QQQ and $4.98 billion for IWM on `new_full` ($4.11 billion and $4.37 billion without same-day expiries), against $16.00 billion for SPY, so no window reaches the $5 billion criterion. This is a consequence of the fixed thresholds, a design decision fixed before the runs, and not a modelling result. The magnitude criterion is the binding one: the persistence and sign-flip tests alone pass 1,428 of 3,878 windows of QQQ (36.8%) and 3,369 of 4,691 of IWM (71.8%), against 1,775 of 4,691 windows of SPY (37.8%). gpt-6-luna labels no window of QQQ or IWM as persistent either (0 of 3,878 and 0 of 4,691 on `new_full`; 0 of 1,700 and 0 of 1,701 on `new_ex0dte`), so the unscaled arms show that the model also rejects every window; they cannot test its discrimination.

The scaled variant multiplies the `new_full` values by one factor per symbol: 4.22 for QQQ (median of 1,699 daily ratios) and 11.43 for IWM (1,700 days). It is a disclosed protocol change (Section 3.6). With it the rule detects 351 of 1,671 windows of QQQ (21.0%) and 902 of 1,672 windows of IWM (53.9%), and gpt-6-luna 361 (21.6%) and 905 (54.1%). The model follows the rule on the scaled inputs: the verdicts agree in 99.4% of the QQQ windows (kappa 0.982) and 99.7% of the IWM windows (kappa 0.994). The paired difference to the unscaled arms is 21.0 percentage points for QQQ (quarter-block bootstrap interval [12.7, 31.2], $p = 5.0 \times 10^{-4}$; exact McNemar test on the non-overlapping subsample $p = 4.9 \times 10^{-4}$) and 53.9 for IWM ([40.7, 67.4], $p = 5.0 \times 10^{-4}$; $p = 4.7 \times 10^{-10}$); for the model the differences are 21.6 and 54.1 percentage points.

The scaled rates depend on the factor and are not a measurement of QQQ or IWM regimes. The ratio of the open-interest notional of SPY to that of the symbol is not constant: for QQQ the daily ratio runs from 0.78 to 38.32 (quartiles 3.30 and 5.73), and for IWM from 1.51 to 110.57 (quartiles 7.73 and 16.46). A single factor puts the exposure on the scale of SPY on a typical day, no more, and another factor would give other rates. The scaled arms therefore show only that the model follows the criteria once the floor is lifted. They show no step after 2023: the rate of the rule on the scaled IWM series is 65.2% in 2020 to 2023 and 35.7% in 2024 and 2025 (shift -29.5 percentage points, $p = 0.061$ by bootstrap, $p = 0.137$ by Fisher exact test on the subsample), and on the scaled QQQ series 25.9% and 13.4% (-12.5 percentage points, $p = 0.170$, $p = 0.292$).

Table 9. Cross-asset replication: windows, typical magnitude and detected windows under the fixed thresholds, by symbol and series. The scaled series are the disclosed protocol change of Section 3.6. The magnitude column is the median over windows of the 30-day mean |GEX| of the series as sent to the model; the scaled series are SPY-scale values, so their magnitudes are not comparable with the unscaled rows of the same symbol. The ex-0DTE arms of QQQ and IWM start in 2020, so gpt-6-luna answered fewer windows than the classifier labelled.

Symbol	Series	Windows	Median mean |GEX| ($B)	Detected (classifier)	Detected (gpt-6-luna)
SPY	`new_full`	4,691	3.86	609 of 4,691 (13.0%)	617 of 4,691 (13.2%)
QQQ	`new_full`	3,878	0.74	0 of 3,878 (0.0%)	0 of 3,878 (0.0%)
IWM	`new_full`	4,691	0.66	0 of 4,691 (0.0%)	0 of 4,691 (0.0%)
SPY	`new_ex0dte`	4,691	3.05	491 of 4,691 (10.5%)	495 of 4,691 (10.6%)
QQQ	`new_ex0dte`	3,878	0.65	0 of 3,878 (0.0%)	0 of 1,700 (0.0%)
IWM	`new_ex0dte`	4,691	0.62	0 of 4,691 (0.0%)	0 of 1,701 (0.0%)
QQQ	`new_full_scaled`	1,671	9.13	351 of 1,671 (21.0%)	361 of 1,671 (21.6%)
IWM	`new_full_scaled`	1,672	11.99	902 of 1,672 (53.9%)	905 of 1,672 (54.1%)



30-day windows ending in the year (QQQ from 2011; ex-0DTE series of QQQ and IWM from 2020). Row 1: lines are the deterministic rule, open markers gpt-6-luna on the same windows; the dotted line is the share of windows that meet the persistence (21 of 30 days) and sign-flip (5) tests, whatever their magnitude; the dotted line and the notes count the complete series with 0DTE. Row 2: median and largest mean |GEX| of the windows ending in the year; the dashed horizontal line is the fixed $5B magnitude threshold. * 2026: windows ending up to 2026-10-09.

Figure 9. Cross-asset replication, SPY, QQQ and IWM (QQQ from 2011). Top row: detection rate by year of the classifier (lines) and gpt-6-luna (markers) on `new_full` and `new_ex0dte`, and the share of windows that pass the persistence and sign-flip tests alone (dotted). Bottom row: yearly median and largest 30-day mean |GEX| against the $5 billion criterion. **Read this figure as:** for QQQ and IWM the detection lines lie at zero in every year while the dotted lines show that many windows pass the persistence and flip tests; the bottom row shows why, since the largest yearly window mean stays below the $5 billion line for QQQ (at most $4.92 billion) and for IWM (at most $4.98 billion). For SPY the largest window mean reaches $16.00 billion.

4.8. Post-Publication Period

Table 10 and Figure 10 cover the post-publication period of Section 3.4: the windows that end in 2026, of which those with ends from 2026-02-13 lie entirely after the end of the published sample (2025-12-31). We use this label rather than out-of-sample, because gpt-6-luna was created on 2026-09-14 and its training cutoff is undocumented. The period has 194 windows with ends from 2026-01-02 to 2026-10-09, of which 165 lie entirely in 2026 (ends from 2026-02-13) and 29 still contain days of 2025. The last window end is 2026-10-09 because equity options moved from Alpha Vantage to Cboe delayed quotes on 2026-10-12 (Section 3.3). On the complete series the rule detects 22 of 194 (11.3%) of the windows that end in 2026; without same-day expiries it detects 41 of 194 (21.1%) (Table 10 also gives the 165 windows entirely in 2026). All detected windows of the complete series are persistent negative-GEX windows, with ends between 2026-03-20 and 2026-04-21. These rates are close to the 12.0% of 2024 and 2025 on the complete series and far from the 100% that the published inputs gave for those years. The Wilson interval for the windows ending in 2026 is 7.6–16.6% (quarter-block bootstrap 0.0–20.3%, 4 blocks). A call-only series would label 194 of the 194 windows as persistent, with a mean |GEX| of $21.74 billion. gpt-6-luna agrees with the rule in 100.0% of the windows on `new_full` and 100.0% on `new_ex0dte` (0 and 0 windows of disagreement, respectively). For QQQ and IWM the

unscaled series give no detection by the rule or by the model (0 of 194 windows each). On the scaled series the rule detects 16.0% (QQQ) and 44.8% (IWM) of the windows ending in 2026 and the model 16.5% and 45.4%. The training data of gpt-6-luna may overlap this period, so the agreement of the model with the rule here shows consistency with the rule and says nothing about forecasting.

Table 10. Post-publication period, SPY, windows ending in 2026 up to 2026-10-09 (those with ends from 2026-02-13 lie entirely in 2026). Cells give windows detected, with the percentage in brackets.

Series	Window set	Windows	Classifier detects (%)	gpt-6-luna detects (%)	Agreement (%)
`new_full`	ends in 2026	194	22 (11.3)	22 (11.3)	100.0
`new_full`	entirely in 2026	165	22 (13.3)	22 (13.3)	100.0
`new_ex0dte`	ends in 2026	194	41 (21.1)	41 (21.1)	100.0
`new_ex0dte`	entirely in 2026	165	41 (24.8)	41 (24.8)	100.0

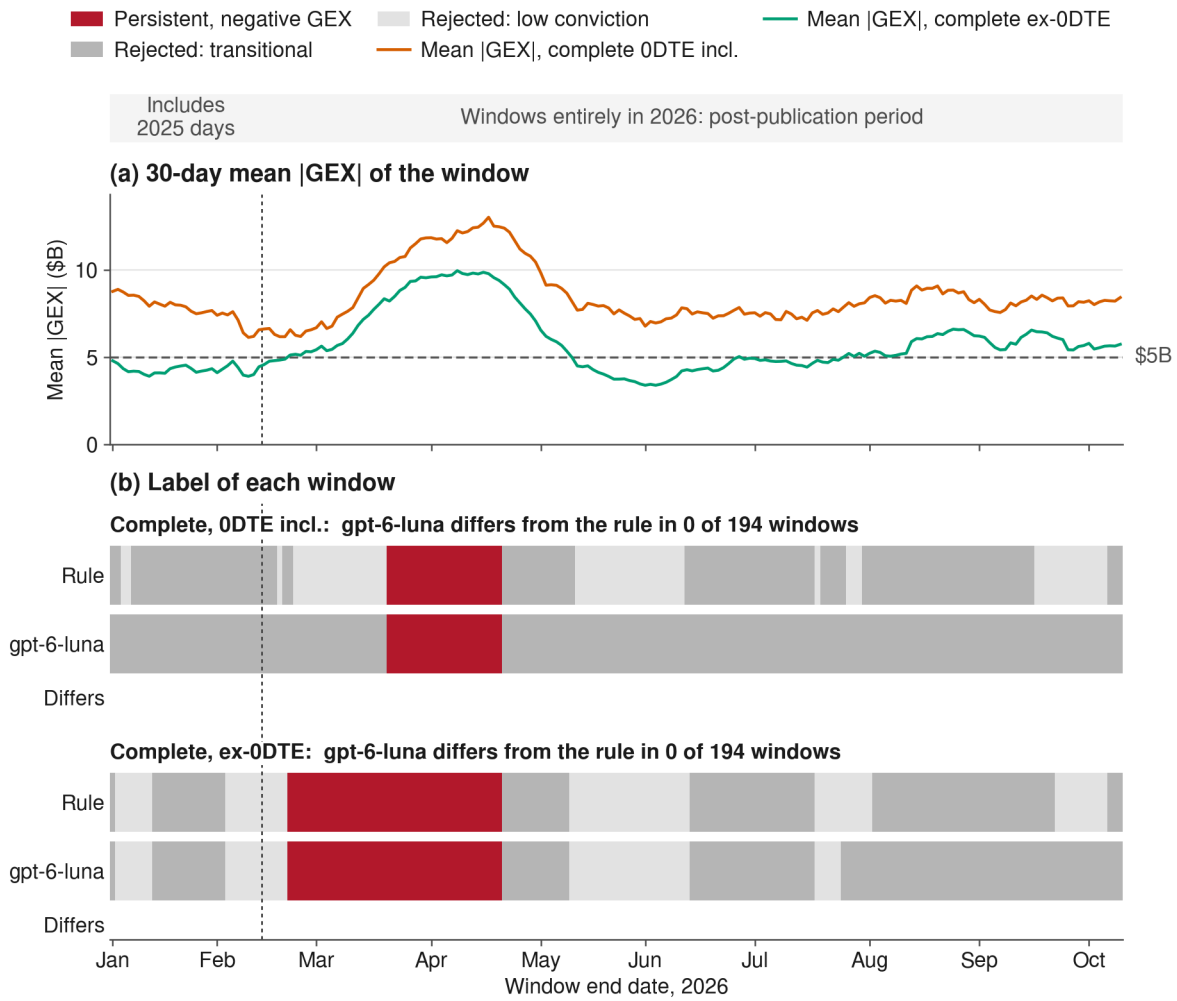


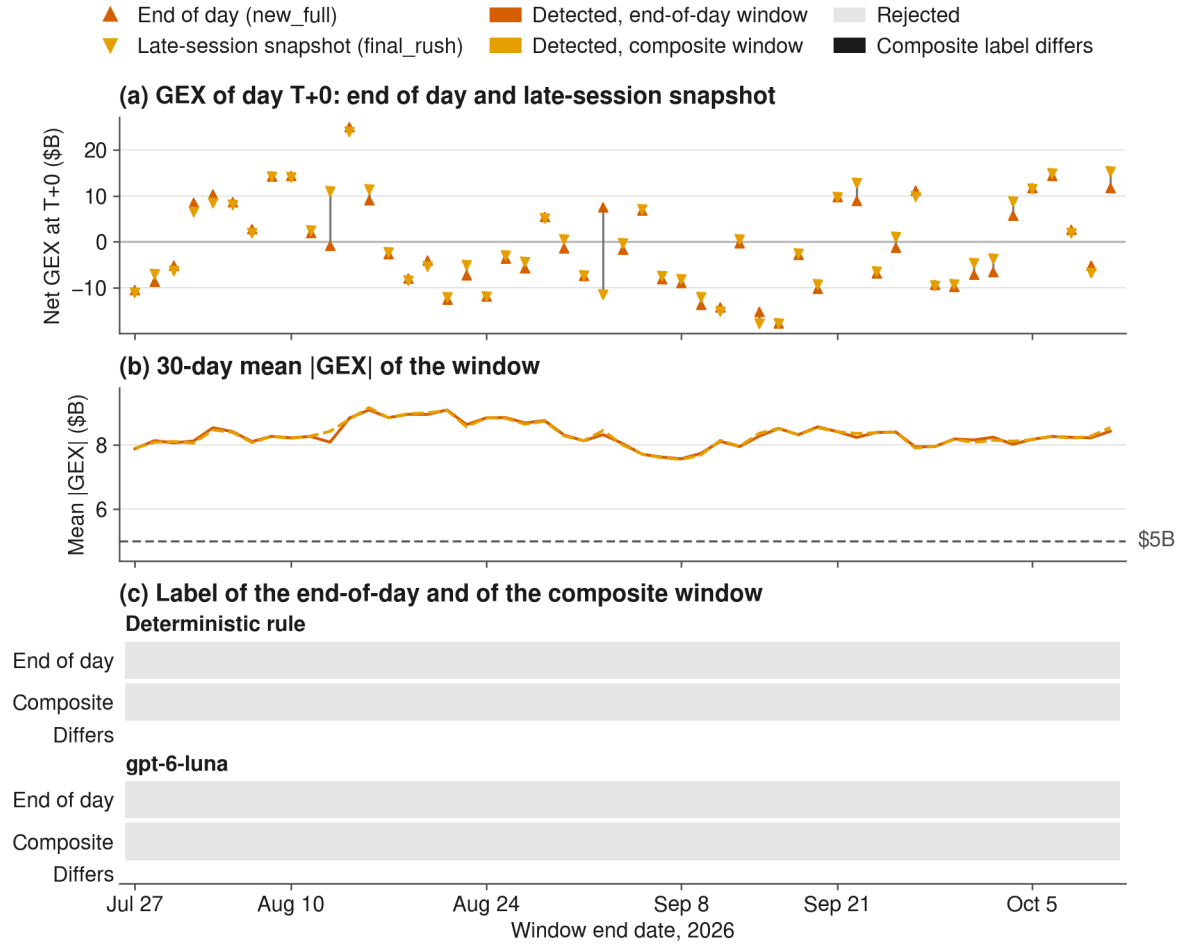
Figure 10. Post-publication period (windows ending in 2026 up to 2026-10-09): per-window classifier and gpt-6-luna labels for SPY under the complete series; the windows that still contain days of 2025 are set apart. **Read this figure as:** the 30-day mean |GEX| of the complete series (a) stays above $5 billion in every window (lowest value $6.13 billion), whereas that of the ex-0DTE series is below it in 85 of 194 windows. The rule labels 22 windows of the complete series and 41 windows of the ex-0DTE series as persistent (red in b), and gpt-6-luna differs from the rule in 0 of the 194 windows of the complete series and in 0 of the 194 windows of the ex-0DTE series.

4.9. Intraday Check

Table 11 and Figure 11 compare the late-session intraday snapshot with the end-of-day value for SPY from 2026-07-27 to 2026-10-09. The snapshot is the `final_rush` slot (19:55:59 UTC), the last independent late-session observation; the `market_close` slot equals the end-of-day row. The irregular days 2026-07-29, 2026-08-03, 2026-09-14 are excluded, which leaves 51 days. The two values of day T+0 have the same sign on 46 of 51 days (90.2%). Their median relative difference is 11.9% and the median absolute difference is \$0.93 billion (mean \$1.69 billion, largest \$19.08 billion). A single day moves the 30-day mean |GEX| of a window only a little: the composite window (the complete series for days $T-29$ to $T-1$ and the snapshot at T+0) differs from the end-of-day window by at most \$0.34 billion. The rule gives the same label for the composite and the end-of-day window on 51 of 51 days, and gpt-6-luna on 51 of 51. This agreement is weak evidence, because no window of the period is persistent: the rule and the model detect 0 and 0 of the 102 windows, and the 30-day mean |GEX| of every window is at least \$7.54 billion, so the verdict depends on persistence and flips, not on the magnitude. With a few dozen days the section reports counts and no p-values.

Table 11. Late-session intraday snapshot versus end-of-day GEX, SPY, 2026-07-27 to 2026-10-09.

Statistic	Value
Days with both observations	51
Days with the same sign (%)	90.2 (46 of 51)
Median relative difference of GEX (%)	11.9
Median absolute difference of GEX (\$B)	0.93
Classifier label agreement of composite and end-of-day windows (%)	100.0 (51 of 51)
gpt-6-luna label agreement of composite and end-of-day windows (%)	100.0 (51 of 51)



SPY, 51 composite window ends. On the 51 regular days the two T+0 values have the same sign on 46; the median relative difference is 11.9%. The rule gives the same label for the composite and the end-of-day window on 51 of 51 days, gpt-6-luna on 51 of 51. Dashed line in (b): the fixed $5B threshold.

Figure 11. Late-session intraday snapshot versus end-of-day GEX, SPY, 2026-07-27 to 2026-10-09. (a) Day T+0 value of the snapshot (`final_rush`) and of the end-of-day row. (b) 30-day mean |GEX| of the composite and of the end-of-day window against the $5 billion criterion. (c) Labels of the classifier and of gpt-6-luna on the two windows. The three irregular days (2026-07-29, 2026-08-03, 2026-09-14) are not in the series. **Read this figure as:** the snapshot and the end-of-day value of day T+0 are close on most days and differ in sign on 5 of 51 days. The two window means in (b) are almost the same line, and in (c) every window is rejected under both labels, with no day on which the composite label differs.

4.10. Model Fidelity to the Rule

Table 12 and Figure 12 compare each model answer with the deterministic classifier on the same window. The verdict of the model equals the verdict of the rule in 99.6% or more of the windows of every SPY arm (99.7% on `full_spy`, 99.7% on `ex0dte_spy`), with kappa between 0.984 and 0.995. The published o4-mini answers equal the rule in all 1,412 windows, and gpt-6-luna reaches 99.4% and 99.7% on the scaled QQQ and IWM series. On the unscaled QQQ and IWM arms the accuracy is 100.0% and kappa is undefined, because the rule labels no window as persistent. Over the answers drawn in panels (c) to (e) of Figure 12 (33,228 parsed answers) the verdict equals the rule's in 99.8% of the 31,752 gpt-6-luna answers and in 100.0% of the 1,476 o4-mini answers.

The reported metrics are less exact than the verdict, and least exact on the unscaled QQQ and IWM arms (82.4% to 89.4%). The share of answers whose reported counts, magnitude and persistence equal the computed values (within 0.5 days, $0.05 billion and 0.5 percentage points) is 99.4% for o4-mini and between 92.7% and 98.1% for gpt-6-luna on the SPY arms. The sign-flip count is exact in 100.0% of the o4-mini answers and in 93.4% to 95.6% of the gpt-6-luna answers on the SPY arms (93.4%

over all gpt-6-luna answers of Figure 12); the model undercounts flips in 6.0% of its answers and overcounts them in 0.6%. Its reported persistence lies within the tolerance of the figure (0.5 percentage points) in 95.0% of the answers and its reported magnitude within $0.05 billion in 98.7%, against 99.9% and 99.6% for o4-mini. The median ratio of reported to computed magnitude is 1.000.

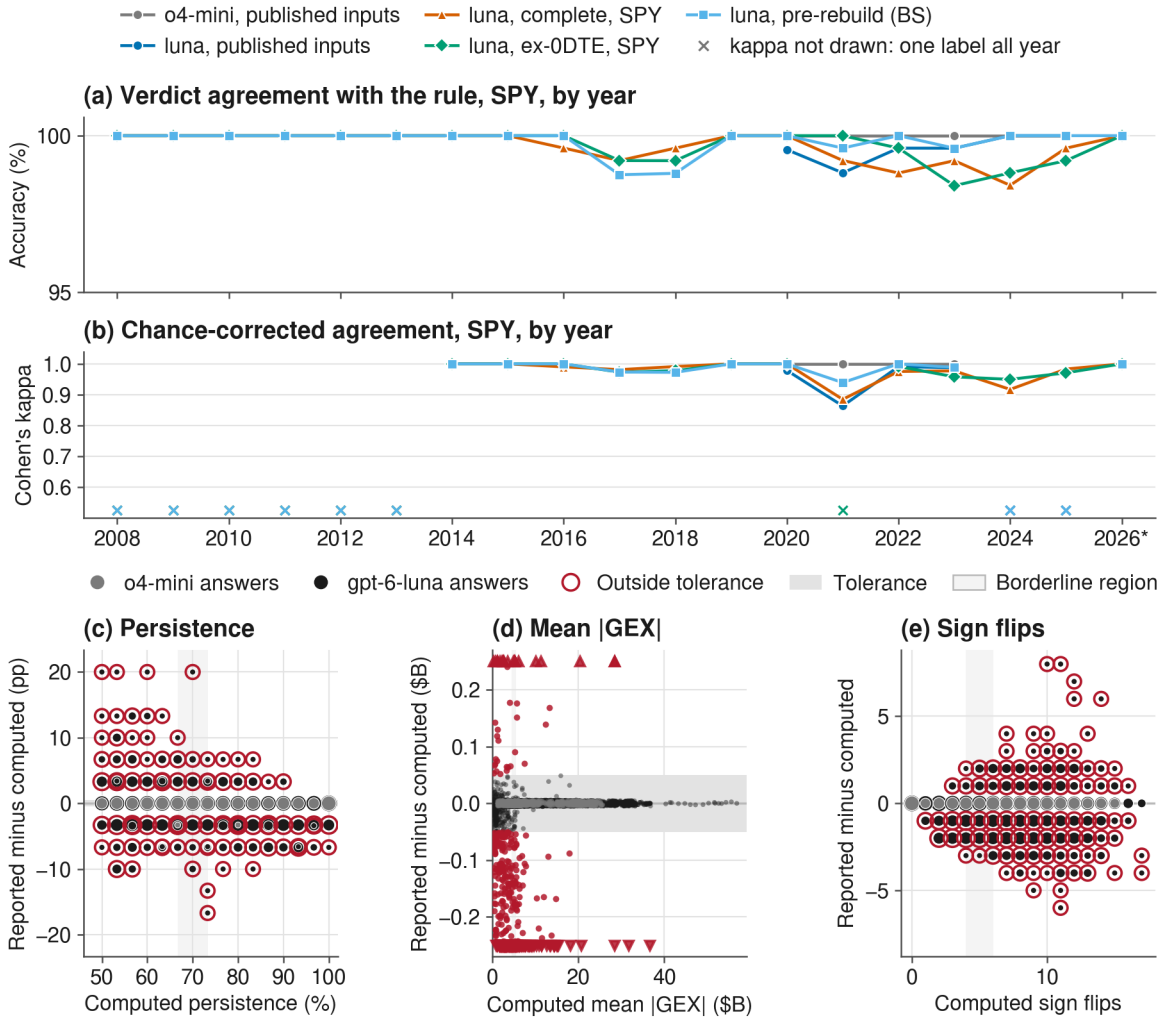
The few verdicts that differ from the rule lie at the thresholds. Seven gpt-6-luna arms have windows in which the verdict differs from the rule (`full_spy`, `ex0dte_spy`, `full_qqq_scaled`, `bridge_old`, `bridge_jan`, `full_iwm_scaled`, `p34_obf_2024`). They hold 60 windows in all, listed in the result files of the repository (Appendix C), with four causes:

- 41 windows in which the model reports at most five sign flips where the rule counts six or more, so that it calls a window persistent that the rule classifies as low conviction;
- 15 windows in which the reported mean magnitude lies on the other side of $5 billion than the computed one;
- 2 windows with overcounted flips;
- 2 windows whose reported values are correct and meet all three criteria but which the model labels transitional (persistence 70.0% and 73.3%, close to the 70% threshold).

In 43 of the 60 windows the model detects a regime that the rule rejects, and in 17 it rejects one that the rule detects. In 52 of the 60 windows the computed values lie in the borderline region of Figure 12, that is a persistence within one day of 21 of 30 days (66.7–73.3%), a mean magnitude between $4.50 and $5.50 billion or 4 to 6 flips. Away from these regions the model reproduces the rule.

Table 12. Fidelity of model answers to the deterministic classifier on the same windows, by arm: accuracy of the binary verdict, Cohen's kappa (– where the classifier or the model labels every window alike), the share of answers whose reported counts, magnitude and persistence equal the computed values (0.5 day, $0.05 billion, 0.5 percentage points), and the share with an exact sign-flip count. The published o4-mini answers of `r1_o4mini`, `r1_p3_o4mini` and `r1_p4_o4mini` form the first block.

Arm	Windows	Accuracy (%)	Cohen's kappa	Metrics agree (%)	Flips exact (%)
`r1_o4mini`	1,412	100.0	1.000	99.4	100.0
`r1_p3_o4mini`	223	100.0	1.000	100.0	100.0
`r1_p4_o4mini`	223	100.0	1.000	100.0	100.0
`bridge_jan`	1,476	99.6	0.992	98.1	95.2
`bridge_old`	4,638	99.8	0.995	94.2	95.6
`full_spy`	4,691	99.7	0.985	92.7	93.4
`ex0dte_spy`	4,691	99.7	0.984	93.5	94.4
`full_qqq`	3,878	100.0	–	82.4	89.7
`full_iwm`	4,691	100.0	–	85.9	92.7
`ex0dte_qqq`	1,700	100.0	–	89.4	92.2
`ex0dte_iwm`	1,701	100.0	–	86.0	89.2
`full_qqq_scaled`	1,671	99.4	0.982	93.5	93.4
`full_iwm_scaled`	1,672	99.7	0.994	97.6	95.9
`p34_obf_2024`	223	99.6	0.985	98.2	96.4
`p34_dated_2024`	223	100.0	1.000	99.6	98.2
`p34_obf_2020`	223	100.0	1.000	100.0	100.0
`p34_dated_2020`	223	100.0	1.000	100.0	100.0
`intraday_spy`	51	100.0	–	96.1	92.2



(c)–(e): 33,228 parsed answers of all registered arms; reported minus the value the rule computes for the same window, against the computed value (marker area grows with the log of the number of identical answers in (c) and (e)). Light band: within one day of the 21-day persistence threshold, within 10% of $5B, within one flip of the limit of 5. Reports inside the tolerance (persistence, mean |GEX|, exact sign flips): o4-mini 99.9%, 99.6%, 100.0%; luna 95.0%, 98.7%, 93.4%. In (d), 197 errors beyond ±$0.25B sit on the frame as triangles. Accuracy on QQQ and IWM: full_qqq 100.0%, ex0dte_qqq 100.0%, full_iwm 100.0%, ex0dte_iwm 100.0%. * 2026: windows ending up to 2026-10-09.

Figure 12. Fidelity of the model answers (o4-mini and gpt-6-luna) to the deterministic classifier. (a), (b) Accuracy and Cohen's kappa of the verdict by series and year. (c)–(e) Reported versus computed persistence, mean |GEX| and sign-flip count, with the borderline region around each threshold shaded. **Read this figure as:** the yearly accuracy stays between 98.4% and 100% for every arm and year (lowest value: `ex0dte_spy` in 2023). In panels (c) to (e) the o4-mini answers (grey) lie on the zero line, whereas the gpt-6-luna answers (black) scatter around it; the errors that matter lie inside the shaded regions, and the largest scatter is in the sign-flip count.

4.11. Dated versus Obfuscated Prompts: The 2024 and 2020 Baselines

Table 13 compares the published o4-mini answers on the inputs of the Phase 3 and 4 baselines, whose prompts carried calendar dates, with gpt-6-luna on the same values with the dates and with relative day labels (Section 3.6). The published o4-mini answers detect 181 of 223 (81.2%) of the 2024 windows and 27 of 223 (12.1%) of the 2020 windows (Wilson 95% intervals 75.5–85.8% and 8.5–17.0%). With the dates, gpt-6-luna reproduces these answers: it gives the same verdict in 223 of 223 windows of 2024 and 223 of 223 of 2020 (kappa 1.000 and 1.000). The deterministic rule on the series as sent gives the same labels as o4-mini in both baselines (100.0% and 100.0% agreement).

Showing the dates changed nothing that the data can detect. With relative day labels gpt-6-luna detects 180 of 223 (80.7%) of the 2024 windows against 181 of 223 (81.2%) with dates, and 27 of

223 (12.1%) of the 2020 windows against 27 of 223 (12.1%). The two labelings differ in 1 window of 2024 and in none of 2020 (paired quarter-block bootstrap interval for 2024 [−1.9, 0.0] percentage points, $p = 0.633$; exact McNemar test $p = 1.000$ at stride one and $p = 1.000$ on the non-overlapping subsample of 8 windows; in 2020 both are $p = 1.000$). The 223 windows of each baseline overlap with stride one, and the quarter-block bootstrap rests on 4 calendar quarters, so these intervals are coarse.

The 2024-versus-2020 contrast of the earlier studies is reproduced under every label type. The difference in detection rate between the two baselines is 69.1 percentage points for the published o4-mini answers ($\varphi = 0.692$; the AIAI paper prints 0.672 and the JRFM article 0.69, and 0.692 is the value recomputed from the 181 of 223 and 27 of 223 counts; Newcombe–Wilson interval [61.6, 74.9]; quarter-block bootstrap interval [28.9, 93.3], $p = 0.003$; Fisher exact test on the non-overlapping subsample of 8 windows per baseline $p = 0.041$), 69.1 for gpt-6-luna with dates ($\varphi = 0.692$) and 68.6 for gpt-6-luna with relative labels ($\varphi = 0.688$; bootstrap $p = 0.005$; Fisher $p = 0.041$). The Fisher test on 8 non-overlapping windows per baseline is far less extreme than the test on all windows, which treats the overlapping windows as independent ($p = 1.8 \times 10^{-52}$).

The contrast is one of magnitudes and not of regimes. The two input series are not on one convention: the 2024 values are a gross magnitude with an imposed negative sign and a mean absolute value of $31.24 billion per day, the 2020 values are a magnitude without sign and have a mean of $2.85 billion (Section 4.1). The 30-day mean |GEX| of the windows averages $30.71 billion in 2024 and $2.56 billion in 2020, against the criterion of $5 billion, so the contrast does not describe a negative against a positive regime, and the type of day label does not change it.

Table 13. Phase 3 (2024) and Phase 4 (2020) baselines: detection rate in percent (detected windows out of 223 in brackets) of the published o4-mini answers, which carried dates, of gpt-6-luna with the dates and with relative day labels (Day T), and of the deterministic classifier on the series as sent. The last column is the exact McNemar p-value of the two gpt-6-luna arms on the non-overlapping subsample (Section 3.8). The last row gives the difference between the two baselines in percentage points with φ in brackets (unpaired).

Baseline	Series	o4-mini, dated	luna, dated	luna, Day T	Classifier	McNemar p
Phase 3, 2024	`p3_2024`	81.2 (181)	81.2 (181)	80.7 (180)	81.2 (181)	1.000
Phase 4, 2020	`p4_2020`	12.1 (27)	12.1 (27)	12.1 (27)	12.1 (27)	1.000
2024 minus 2020, pp (φ)		69.1 (0.692)	69.1 (0.692)	68.6 (0.688)	69.1 (0.692)	–

4.12. Open-Weights Models on Local Hardware

The seven open-weights arms repeat the bridge, the two complete SPY arms and the four baseline arms with gpt-oss-20b (Section 3.5). They sent 12,398 requests, the 40-window pilot, the 5% repeats and the retries included, in 10.0 GPU-hours on the shared A100 GPUs, at no cost. Fourteen first-pass answers ended at the limit of 4,000 completion tokens; each was requested again at 8,000 tokens, and every retry parsed. Of the 11,750 windows of the seven arms, 11,748 hold a parsed answer. The two that do not are windows of `oss_full_spy` whose answer closes the reasoning field with a typographic quotation mark followed by control tokens of the model, so that no parser recovers the JSON object; one answer of `oss_bridge_jan` with the same defect was recovered by the fallback parser. The answers of gpt-oss-20b are longer: between 1,126 and 1,995 completion tokens per arm on average, against 736 to 788 for gpt-6-luna on the bridge and the complete series.

On the published inputs gpt-oss-20b gives the verdict of o4-mini in 1,406 of the 1,412 published windows (99.6%, kappa 0.991), as gpt-6-luna does in 1,407 (99.6%, kappa 0.993). Like gpt-6-luna it detects every window that o4-mini detects, and it adds 6 detections where gpt-6-luna adds 5 (exact McNemar test at stride one $p = 0.031$). It detects all 501 windows of 2024 and 2025 on these inputs, so the open-weights model reproduces the published saturation from the published inputs, with a rise of 82.4 percentage points from 2020–2023 to 2024–2025 (quarter-block bootstrap $p = 0.0005$).

On the complete series gpt-oss-20b follows the rule as closely as gpt-6-luna does. It agrees with the rule in 99.7% of the 4,689 parsed windows of `oss_full_spy` (kappa 0.988, against 0.985 for gpt-6-luna) and in 99.9% of the 4,691 windows of `oss_ex0dte_spy` (kappa 0.992, against 0.984). For 2024 and 2025 it detects 62 of 502 windows (12.4%; quarter-block bootstrap 5.7–19.8%), between the rule (60, 12.0%) and gpt-6-luna (63, 12.5%), and its change from 2020–2023 to 2024–2025 is -10.1 percentage points (bootstrap $p = 0.103$), against -10.2 for the rule ($p = 0.108$). Without same-day expiries it detects 75 of 502 windows of 2024 and 2025 (14.9%), against 14.7% for the rule and 15.3% for gpt-6-luna, so for this model, too, removing them does not restore the published rates.

The two current models give the same verdict in 99.4% of the windows of the complete series (4,660 of 4,689; kappa 0.973) and in 99.6% without same-day expiries (4,670 of 4,691; kappa 0.976). Where they differ, the rule sides with gpt-oss-20b in 16 and with gpt-6-luna in 13 of the 29 windows of `new_full`, and in 14 and 7 of the 21 windows of `new_ex0dte`, so gpt-oss-20b follows the criteria at least as closely as gpt-6-luna. On the windows that both answered, the two models detect almost the same number (gpt-6-luna 616 and gpt-oss-20b 617 on `new_full`, 495 and 496 on `new_ex0dte`; exact McNemar test $p = 1.000$ in both). On the published inputs they agree in 99.2% of the 1,476 windows (kappa 0.984), with 6 of the 12 differences resolved in favour of each.

For gpt-oss-20b as for gpt-6-luna, the dates in the prompt changed nothing that the data can detect. With relative labels it detects 185 of 223 windows of the 2024 baseline (83.0%) and with the dates 183 of 223 (82.1%); the two labelings differ in 6 windows, 4 detected with relative labels only and 2 with dates only (paired quarter-block bootstrap interval [0.0, 2.4] percentage points, $p = 0.633$; exact McNemar test $p = 0.688$ at stride one and $p = 1.000$ on the non-overlapping subsample of 8 windows), and in 2020 both labelings detect 27 of 223. With the dates it differs from the published o4-mini answers in 2 windows of 2024, both detected by gpt-oss-20b only, and in none of 2020. The 2024-versus-2020 contrast is 70.9 percentage points with relative labels ($\varphi = 0.709$) and 70.0 with dates ($\varphi = 0.701$), with the Fisher test on the non-overlapping subsample at $p = 0.041$ in both cases. The 2024 baseline with relative labels is the arm where gpt-oss-20b departs most from the rule (185 detections against 181; agreement 98.2%, kappa 0.939).

Repeated requests give the first verdict again in 74 of 74 bridge windows, 234 of 235 windows of the complete series, 241 of 241 windows without same-day expiries and 11 of 11 in each baseline arm. The reported metrics match the computed ones about as well as those of gpt-6-luna: on the complete series the reported mean |GEX| lies within 5% of the computed value in 99.5% of the answers (gpt-6-luna 98.7%), and the sign-flip count is exact in 93.4% (gpt-6-luna 93.4%). A model whose weights are fixed at one revision therefore reproduces the bridge, the rates of the complete series and the absence of a date effect, so these results do not depend on the hosted alias.

Llama 3.1 8B answers the same seven arms with 12,378 requests, the pilot and the 5% repeats included, in 3.3 GPU-hours. Every answer parses at the first attempt and none comes near the token limit (about 170 completion tokens on average), but the verdicts carry no information about the window: the model detects a regime in all 1,476 windows of the bridge, in all 223 windows of each baseline arm and in 4,690 of the 4,691 windows of each complete series, the exception being the same window (ending 2010-01-28) in both. Its agreement with the rule therefore equals the detection rate of the rule (45.2% on the bridge, 13.0% and 10.5% on the complete series, 81.2% and 12.1% on the baselines), with kappa 0.000 in every arm, and repeated requests return the first verdict in all 588 repeated windows. On the published inputs it detects all 501 windows of 2024 and 2025, as o4-mini did, but it also detects all 223 windows of the 2020 baseline, where o4-mini detected 27, so its 2024-versus-2020 contrast is zero. A detection rate of 100% in 2024 and 2025 is thus what a constant answer gives; the contrast with 2020 is the part of the published test that a model has to pass, and Llama 3.1 8B does not pass it.

The metrics that Llama 3.1 8B reports do not follow the inputs either. Its reported mean |GEX| lies within 5% of the computed value in 14.3% to 64.1% of the answers per arm (gpt-6-luna: at least 98.4%), and its sign-flip count is exact in 2.7% to 13.9%. The schema puts the verdict first, so this model, which

has no reasoning phase, writes the verdict before any working, and the reasoning field then argues for the verdict with comparisons that are false: for the window of `new_full` ending 2008-02-27 it writes that "the persistence percentage of 66.67% meets the 70% threshold" and that "the average magnitude of $2.45B is greater than the $5B threshold". A model of this size without a reasoning phase therefore does not perform the task, and this arm cannot test whether a model of a second developer follows the rule.

Table 14. The open-weights arms: detection rate in percent (detected windows in brackets) of the deterministic rule, of gpt-6-luna and of gpt-oss-20b on the same windows, and the agreement of gpt-oss-20b with gpt-6-luna and with the rule in percent (Cohen's kappa in brackets). Windows are those with a parsed answer of gpt-oss-20b; the rule and gpt-6-luna rates are over all windows of the arm.

Arm	Windows	Rule	gpt-6-luna	gpt-oss-20b	vs gpt-6-luna	vs rule
`oss_bridge_jan`	1,476	45.2 (667)	45.5 (671)	45.6 (673)	99.2 (0.984)	99.6 (0.992)
`oss_full_spy`	4,689	13.0 (609)	13.2 (617)	13.2 (617)	99.4 (0.973)	99.7 (0.988)
`oss_ex0dte_spy`	4,691	10.5 (491)	10.6 (495)	10.6 (496)	99.6 (0.976)	99.9 (0.992)
`oss_p34_obf_2024`	223	81.2 (181)	80.7 (180)	83.0 (185)	97.8 (0.925)	98.2 (0.939)
`oss_p34_dated_2024`	223	81.2 (181)	81.2 (181)	82.1 (183)	99.1 (0.970)	99.1 (0.970)
`oss_p34_obf_2020`	223	12.1 (27)	12.1 (27)	12.1 (27)	100.0 (1.000)	100.0 (1.000)
`oss_p34_dated_2020`	223	12.1 (27)	12.1 (27)	12.1 (27)	100.0 (1.000)	100.0 (1.000)

5. Discussion

5.1. What Survives

The re-run leaves intact the finding that detection is rare before 2024 and varies from year to year. In 2021 to 2023 the rule gives 4.0%, 32.3% and 19.3% on the published inputs and 3.6%, 40.6% and 22.4% on the complete series, and the model reproduces the rule on both. Before 2020 the complete series detects 10.2% of the windows pooled over 2008–2019 and none up to 2013, where no window reaches the magnitude criterion (Section 4.6), so low detection rates are not specific to the years after 2020. Without the magnitude criterion the share of windows with a persistent sign falls from 52.1% in 2008 to 2013 to 12.0% in 2024 and 2025, so neither reading shows a rise after 2023. It does not reproduce the saturation of 2024 and 2025: the published inputs give 100.0% in both years and the complete series 10.3% and 13.6% (Section 4.2). Nor does it reproduce their sign: every published detection of 2024 and 2025 is a positive regime, whereas half of the 60 detections of the complete series are negative and end in March and April 2025. The period contrast of 2020 to 2023 against 2024 and 2025 is a rise of 83.0 percentage points on the published inputs and a change of -10.2 points on the complete series, a change that the data do not distinguish from zero (bootstrap $p = 0.108$). The lineage finding bears on the 0DTE reading of the 2023-to-2024 shift. Same-day expiries carry 45.9% of the absolute net GEX of SPY in 2024, against 22.1% in 2019, so the growth of 0DTE trading is visible in the exposure itself. The sudden step of the detection rate between 2023 and 2024 is not: it coincides with the change of the input from the complete series to the call leg (Section 4.1), and it does not appear on the complete series. Removing same-day expiries does not restore it (12.0% with and 14.7% without them in 2024 and 2025; Section 4.5). We therefore read the published rise as a property of the published inputs. The complete series supports only the weaker statement that same-day expiries form a large part of the exposure, and it claims no mechanism that links them to the regime labels. The years of the replay are not independent observations, so these readings rest on the overlap-aware checks of Section 3.8 and not on the Wilson intervals.

5.2. What the Model Tracks

The verdict of the model follows the numbers it is shown and the criteria in the prompt. In the full runs it equals the rule in 99.7% of the windows of both complete SPY arms (kappa 0.985 and

0.984). The few verdicts that differ lie at the thresholds: 60 windows over seven arms, 52 of them in the borderline region of persistence, magnitude or flip count (Section 4.10). Obfuscation therefore validates the model on a given series. It does not show that the series is what the prompt says it is. The same models, prompts and rule give 100.0% or about 12% on the same years, depending only on which series enters the prompt, so the provenance of the input series is part of the test. The 69.1-point contrast between the 2024 and 2020 baselines is reproduced with relative labels, but those two inputs differ in convention and scale (mean |GEX| of \$30.71 and \$2.56 billion), so it does not describe a negative against a positive regime (Section 4.11). The rerun of the Phase 3 and 4 baselines points the same way for the visibility of dates. With calendar dates gpt-6-luna reproduces the November 2025 answers in 223 of 223 windows in each baseline, and with relative day labels it differs in 1 window of 2024 and in none of 2020 (Section 4.11). Visible dates changed nothing that these data can detect. The model is sensitive to the values and the criteria, and it shows no sensitivity to dates in the prompt that the data can detect (1 of 223 windows differs in the 2024 baseline). For labelling, the rule is the better instrument on this task: it is exact by definition, it needs no API call, and the verdicts of the model add nothing to it, while the model undercounts sign flips in 6.0% of its answers (Section 4.10). The model is the object of the test and not a replacement for the rule. The open-weights gpt-oss-20b reads the same series in the same way: it agrees with the rule in 99.7% of the windows of the complete series, and where it and gpt-6-luna differ, the rule sides with gpt-oss-20b at least as often as with gpt-6-luna (Section 4.12). What the design measures is what its verdict depends on: the values and the criteria, and neither the dates in the prompt (Section 4.11) nor the change of model on fixed inputs (Section 4.3).

5.3. Threats to Validity

All model arms use models of one developer, OpenAI. The alias gpt-6-luna has no dated snapshot, and we store the creation time of the model object and the response identifier of every answer instead; the open-weights gpt-oss-20b is fixed at one revision and can be run again (Section 4.12). The open-weights arms remove the dependence on the alias, not on the developer, so the agreement with the rule may differ for models of other developers. Equity options moved from Alpha Vantage to Cboe delayed quotes on 2026-10-12, so the last window ends on 2026-10-09, and the end-of-day open interest differs from the intraday value (the sign of the net GEX is the same on 46 of 51 days, 90.2%, and the check rests on a single snapshot type, the `final_rush` slot). The prompt shows values rounded to 0.01B whereas the rule uses the unrounded values, which can move a window across a threshold. Windows have stride one and share 29 of their 30 days, so the Wilson intervals are optimistic, and we rely on the quarter-block bootstrap, which has few blocks per year, and on the non-overlapping subsample, which has few windows (Section 3.8). Four trading days stay absent from the daily table because no chain exists for them (2018-11-06, 2019-09-03 and 2020-06-23 for all three symbols, and 2024-09-30 for QQQ), so 87 of the 4,691 windows of SPY and of IWM and 116 of the 3,878 windows of QQQ span a missing NYSE trading day (Table 2). They are kept in every rate. Without them the rule detects 12.8% instead of 13.0% of the windows of `new_full` for SPY (-0.15 percentage points) and 10.4% instead of 10.5% of those of `new_ex0dte` (-0.11), and still none of the windows of QQQ and IWM. On the scaled series the rate rises by 0.76 percentage points for QQQ (21.0% to 21.8%, 58 windows) and by 0.95 for IWM (53.9% to 54.9%, 29 windows), and on the published inputs by 0.74 (45.2% to 45.9%, 87 windows). The largest change is on the 2020 baseline, where 29 of 223 windows span 2020-06-23 and the rule rate moves from 12.1% to 13.9% (1.81 percentage points). No window of SPY that ends in 2024 or 2025 spans a missing day, so the rates of those years are unaffected. The rates of gpt-6-luna move by similar amounts (at most 0.96 percentage points on the arms of Table 3 other than the 2020 baseline and the 40-window pilot (1.28), and 1.81 on the 2020 baseline arms). The thresholds come from the earlier work and are fixed in dollars for every symbol; for QQQ and IWM they produce a floor effect (no window is detected in 3,878 and 4,691 windows) that the unscaled arms cannot test. They are also fixed over time, although GEX grows with the square of the price and with open interest, so the magnitude criterion binds in the early years (no SPY window reaches it before

2014, and no window of `new_ex0dte` in 2021) and the rates of different years do not lie on one real scale; Section 4.6 gives the shares without that criterion. Relative day labels remove the dates but not every clue to the period: the 30-day mean |GEX| of the SPY windows averages $1.16 billion in 2010 and $6.48 billion in 2024 on `new_full`, so the scale of a window can indicate its era. The dated and the relative labels give the same verdict in 445 of the 446 windows of the two baselines (Section 4.11), and the verdicts of the model agree with the rule, which reads no dates, in 99.7% of the windows of `full_spy`, so an inference of the period from the scale could change only a small share of the labels. GEX models dealer positioning and does not observe it: Equation (1) assigns the sign by option type because open interest does not record which side dealers hold, so every label of this study inherits that assignment, and data that identify the dealer side could give other signs. Each window is answered once unless an arm was repeated; the repeat of the bridge arm differs from the first pass in 11 windows (99.3% agreement), so a single run carries a small amount of sampling noise at the thresholds. The model undercounts sign flips in 6.0% of its answers, which is the main source of the verdicts that differ from the rule, and the stored count of the rule is the reference. The post-publication period is short (194 windows ending in 2026, of which 165 lie entirely after 2025) and is not out-of-sample for the model, whose training data may cover it; the agreement there shows consistency with the rule and no predictive skill. The study tests the labels and not their consequences for prices: it does not test whether the windows labelled persistent differ in realised volatility or returns. Such a test needs a design of its own, because the exposure depends on the price path (gamma is largest near the money, so a fall of the underlying raises the weight of the puts in Equation (1)), and the sign of GEX is in part a consequence of the returns that it would be used to explain. The raw option chains are not deposited because of their size; the tables deposited at Zenodo, which hold the inputs, the labels and the answers, are the basis for reproduction (Section 3.7). The Phase 3 and 4 baselines are replayed verbatim, including two sign conventions and ten placeholder rows, and the scaled QQQ and IWM variants use one factor per symbol, a disclosed protocol change that was decided after the thresholds were fixed; their rates depend on the factor and do not measure QQQ or IWM regimes.

5.4. Implications

The study suggests four practices for the validation of AI-enabled models. Inputs should be versioned and stored with each model run, together with the labels and the raw answers, so that every verdict can be traced to its values. A lineage check between runs should compare the inputs of a new run with those of the previous run before the labels are compared. For a net exposure such as GEX the check can start with the legs: the call leg alone is positive by construction and meets the persistence and stability criteria in every window (Section 4.2), and the published inputs were positive on each of the 565 trading days from 2023-09-29 to 2025-12-31, whereas no year of the complete series of any symbol keeps one sign. A comparison of the net series with each leg would have shown the missing put rows before any model was run. Drift alarms on upstream rebuilds should flag a change in the construction of an input table. A bridge run on identical inputs should accompany the retirement of a model, as the bridge between o4-mini and gpt-6-luna does here (kappa 0.993 on 1,412 windows), and a model with open weights at a fixed revision can carry the bridge beyond the life of a hosted alias (gpt-oss-20b: kappa 0.991 against o4-mini on the same windows; Section 4.12). These practices address the theme of the Special Issue, timely AI advances and their interdisciplinary application, at the point where an upstream data change reaches the model without any change to the model.

6. Conclusions

This study re-runs the detection of persistent dealer-gamma regimes on the complete data. On the published inputs the rule detects 100.0% of the windows of 2024 and 2025, and on the complete series it detects 12.0% (60 of 502; quarter-block bootstrap 5.1–19.6%). The regime labels are therefore sensitive to the input convention, and the earlier saturation is a property of the inputs; the complete series shows no increase after 2023 that the data distinguish from zero. A second model, gpt-6-luna, agrees

with the published o4-mini answers on 99.6% of 1,412 windows (kappa 0.993), follows the rule on the complete series in 99.7% of the windows, and shows no sensitivity to dates in the prompt that the data can detect (1 of 223 windows differs in the 2024 baseline). An open-weights model, gpt-oss-20b, run locally at a fixed revision, gives the same picture: it agrees with gpt-6-luna in 99.4% and with the rule in 99.7% of the windows of the complete series, detects 12.4% of the windows of 2024 and 2025, and shows no date effect that the data can detect either. Same-day expiries carry 41–46% of the absolute net GEX in 2024–2026 but their removal does not restore the published rates (14.7% against 12.0%); the post-publication windows of 2026 (11.3%) and the late-session snapshot (51 of 51 labels unchanged) give the same picture. The scope is limited to models of one developer (an API alias and one open-weights model), dollar thresholds fixed across symbols and over time, a sign convention that imputes dealer positions and overlapping windows. The scaled QQQ and IWM variants and the single intraday snapshot are checks and not measurements. Future work includes a variant with symbol-specific and price-adjusted thresholds in the prompt (a new prompt version, unlike the scaled inputs used here), intraday open interest, a test of the price behaviour of the labelled windows that accounts for the dependence of GEX on the price path, models of other developers, and standards for versioned inputs in LLM validation studies.

Appendix A. Prompt, Schema and API Configuration

This appendix reproduces the complete request sent to the model for each 30-day window, so that the experiment can be repeated from the manuscript alone. The prompt text is the output of `MechanicsPromptBuilder.build_regime_prompt` (file `src/llm/mechanics_prompt_builder.py`); the schema and the request body are the constant `REGIME_JSON_SCHEMA` and the function `chat_body` of `scripts/validation/electronics/common.py`. The listings were generated from these objects and compared with them.

Appendix A.1. Model and API Configuration

- **Model:** `gpt-6-luna`, an OpenAI model alias without a dated snapshot. The creation timestamp of the model object (2026-09-14) is stored with every run and the response identifier with every answer. The published run used `o4-mini-2025-04-16`.
- **Access:** OpenAI Batch API, endpoint `/v1/chat/completions`, completion window of 24 hours, at most 1,500 requests per batch job. Each input line is one request that carries the key `<arm>|<symbol>|<window_end>` in `custom_id`; a test-retest duplicate adds the suffix `#rep1`, and its arm is stored as `<arm>#rep1`.
- **Reasoning effort:** `medium`, the default effort of o4-mini in the published run.
- **Maximum completion tokens:** 4,000; reasoning tokens count against the limit. An answer that ends with `finish_reason` = `length` is requested once more with 8,000.
- **Response format:** the strict JSON schema of Appendix A.3. The earlier runs requested the same nine fields in the prompt text only; this is the one protocol change of the study.
- **Not sent:** temperature, seed, system message, tools and stop sequences.
- **Parsing:** an answer is parsed as JSON; a second attempt tolerates raw control characters inside strings, and a last fallback extracts the JSON object from free text. The parse status of every answer (`ok`, `fallback`, `failed` or `missing`) is stored.
- **Prices:** USD 0.05 per million input tokens and USD 0.25 per million output tokens on Batch, twice that on the standard API.
- **Open-weights arms (Section 4.12):** `gpt-oss-20b` from the repository `openai/gpt-oss-20b` at revision `6cee5e81ee83917806bbde320786a8fb61efebee`, read offline from a local copy, answered by the offline batch runner of vLLM 0.32.0 on one NVIDIA A100-SXM4-80GB GPU per job. The engine uses 22% of the GPU memory, a maximum model length of 8,192 tokens (12,096 for a retry at 8,000 completion tokens), engine seed 261, its structured-output decoding for the JSON schema with the reasoning parser of gpt-oss, and the PyTorch sampler (`VLLM_USE_FLASHINFER_SAMPLER=0`). The request body is that of Appendix A.4 with the model name `gpt-oss-20b`;

no price applies. `llama-3.1-8b-instruct` comes from `meta-llama/Llama-3.1-8B-Instruct` at revision `0e9e39f249a16976918f6564b8830bc894c89659` in bfloat16, with the same engine settings except 22% of the GPU memory (27% for its pilot) and no reasoning parser; its request body omits `reasoning_effort`, and sampling follows the repository's `generation_config.json` (temperature 0.6, top-p 0.9). `qwen3-8b` comes from `Qwen/Qwen3-8B` at revision `b968826d9c46dd6066d109eabc6255188de91218` in bfloat16, in thinking mode, with the reasoning parser of Qwen3 and 27% of the GPU memory (up to 45% when more was free); its request body is that of gpt-6-luna with the model name `qwen3-8b`, and sampling follows the repository's `generation_config.json` (temperature 0.6, top-p 0.95, top-k 20).

Identical requests need not return identical answers, because no seed is set (and, for the open-weights arms, because the batches that share the engine vary). The test-retest duplicates of Section 3.6 measure this variation, and the deterministic rule, not the model, supplies the reference label of each window.

Appendix A.2. Prompt Text

The prompt is a single user message. The placeholder `{gex_data_table}` stands for the 30 data lines of a window, one per day, in the form `Day T-29: +3.42B` to `Day T+0: -12.18B`: a relative day label, the net GEX in billions of US dollars with two decimals and an explicit sign, and neither a date nor a ticker. In the two dated arms (`p34_dated_2024` and `p34_dated_2020`) the label is instead the ISO date of the trading day, as in the November 2025 runs (`2024-01-02: -33.01B`). The rest of the text is the same for every window, series and symbol (prompt version `v2.0_postgresql`) and is the text of the Phase 4A run.

The listing was checked against the stored requests. With the 30 data lines replaced by the placeholder, it equals the user message of 35,365 of 35,365 requests (the 35,833 requests of Table 3 less the 468 requests of the two dated arms). Each stored answer also carries the sha256 of the prompt that produced it in `detection_metadata.prompt_sha256`.

```
You are a market structure analyst specializing in dealer gamma positioning regimes.

TASK: Analyze this 30-day period and determine if it represents a PERSISTENT regime where dealer
↪ constraints create forced, directional flows.

## 30-DAY GEX DATA

{gex_data_table}

## REGIME CLASSIFICATION FRAMEWORK

### PERSISTENT REGIMES (Detect These)

**1. PERSISTENT POSITIVE REGIME**
- Definition: Dealers are LONG gamma, forced to sell into strength
- Criteria:
  * >70% of days (21+/30) have positive net GEX
  * Average magnitude >$5B
  * ≤5 sign flips across 30 days
  * Stable directional constraint

**Mechanism**: When dealers hold long gamma:
- Price rises → Dealers MUST sell shares (rebalance)
- Price falls → Dealers MUST buy shares (rebalance)
- Creates dampening, mean-reverting flows
- Constraint is STRUCTURAL (dealers cannot avoid)

**2. PERSISTENT NEGATIVE REGIME**
- Definition: Dealers are SHORT gamma, forced to buy into strength
- Criteria:
  * >70% of days (21+/30) have negative net GEX
  * Average magnitude >$5B
  * ≤5 sign flips across 30 days
  * Stable directional constraint
```

```
**Mechanism**: When dealers hold short gamma:
- Price rises → Dealers MUST buy shares (chase)
- Price falls → Dealers MUST sell shares (chase)
- Creates amplifying, momentum flows
- Constraint is STRUCTURAL (dealers cannot avoid)

---

### NON-REGIMES (Reject These)

**3. TRANSITIONAL (Reject)**
- Frequent sign flips between positive/negative GEX
- No dominant regime direction (less than 70% same sign)
- Market in regime change period
- Example: 15 positive days, 15 negative days (50/50 split)

**Why Reject**: No persistent constraint. Dealers face mixed conditions daily. Not a structural
↪ regime.

**4. LOW CONVICTION (Reject)**
- Consistent sign BUT weak magnitude (<$5B average)
- Example: 25 days positive, avg $2B GEX
- Insufficient constraint to create persistent forced flows

**Why Reject**: Even if sign is consistent, magnitude too weak to force dealers into meaningful
↪ positions. Not a structural constraint.

---

## ANALYSIS QUESTIONS

Systematically evaluate the 30-day window:

**Step 1: Sign Persistence**
1. Count days with positive net GEX
2. Count days with negative net GEX
3. Calculate persistence percentage: max(positive_days, negative_days) / 30 * 100
4. Does it meet 70% threshold (21+ days)?

**Step 2: Magnitude Assessment**
1. Calculate average GEX magnitude (absolute value): sum(|net_gex|) / 30
2. Is average magnitude ≥$5B?
3. Check for extreme outliers that might distort average

**Step 3: Stability Check**
1. Count sign flips: How many times does GEX switch from pos→neg or neg→pos?
2. Are there ≤5 sign flips across 30 days?
3. Stable regime should have low flip count

**Step 4: Regime Classification**
- If Steps 1, 2, 3 all pass AND positive dominates → PERSISTENT POSITIVE
- If Steps 1, 2, 3 all pass AND negative dominates → PERSISTENT NEGATIVE
- If Step 1 passes but Step 2 fails → LOW CONVICTION (reject)
- If Step 1 fails → TRANSITIONAL (reject)

---

## CONFIDENCE CALIBRATION (Mechanical Guidance)

Use these concrete anchors to calibrate confidence:

**90-100 (Very High Confidence)**
- 25-30 days same sign (83-100% persistence)
- Average magnitude >$10B
- 0-2 sign flips (highly stable)
- Example: "29 negative days, avg $15B, 1 flip"

**70-89 (High Confidence)**
- 21-24 days same sign (70-80% persistence)
- Average magnitude $5-10B
- 2-4 sign flips (moderately stable)
- Example: "23 negative days, avg $7B, 3 flips"

**50-69 (Borderline - Use with Caution)**
- 18-20 days same sign (60-67% persistence)
```

```
- Average magnitude $3-5B
- 5-7 sign flips
- Example: "20 negative days, avg $4B, 6 flips"
- **Note**: Borderline cases should generally be REJECTED unless other factors strengthen confidence

**0-49 (Reject - Not Persistent)**
- <18 days same sign (<60% persistence)
- OR average magnitude <$3B
- OR >7 sign flips
- These are NOT persistent regimes

**Important**: Confidence is a FILTER, not a probability. Use it to distinguish clear regimes (70+) from borderline (50-69) from noise (<50).

---

## OUTPUT FORMAT (JSON)

Provide your analysis in this exact JSON structure:

```json
{
    "regime_detected": true/false,
    "regime_type": "persistent_positive|persistent_negative|transitional|low_conviction",
    "positive_days": <count as integer>,
    "negative_days": <count as integer>,
    "avg_magnitude_billions": <value as number>,
    "sign_flips": <count as integer>,
    "persistence_pct": <percentage as number>,
    "confidence": <integer 0-100>,
    "reasoning": "Explain step-by-step why this is/isn't a persistent regime. Reference specific metrics (persistence %, avg magnitude, sign flips). If rejecting, state which criterion failed."
}
```

IMPORTANT: All numeric fields (confidence, positive_days, negative_days, sign_flips, avg_magnitude_billions, persistence_pct) MUST be numbers (integers or decimals), NOT words like "thirty-five" or "fifty".

regime_detected Rules:
- `true` ONLY if regime_type is "persistent_positive" or "persistent_negative"
- `false` if regime_type is "transitional" or "low_conviction"

KEY PRINCIPLES

1. **Selectivity is Expected**: Most windows will NOT be persistent regimes (expect 30-50% detection rate)

2. **ALL Criteria Must Pass**: Persistence + Magnitude + Stability required for detection

3. **Rejection is Valid**: Saying "no persistent regime" is a correct answer for transitional/weak periods

4. **Mechanical Over Qualitative**: Use concrete thresholds (70%, $5B, 5 flips) rather than subjective judgment

5. **Structural Focus**: Only detect when dealers are FORCED into directional positions by constraints

Analyze the 30-day GEX data above and provide your regime classification in JSON format.
```

Appendix A.3. Output Schema

The answer is constrained by the following strict JSON schema, passed as `response_format`. With `strict` set, the model can neither omit nor add a field. The boolean `regime_detected` is the primary label of the study and `regime_type` the secondary one; the counts, the mean magnitude and the persistence are the model's own report and are compared with the values of the rule (Section 4.10).

```
{
  "type": "json_schema",
  "json_schema": {
    "name": "regime_classification",
    "strict": true,
    "schema": {
      "type": "object",
      "additionalProperties": false,
      "properties": {
        "regime_detected": {
          "type": "boolean"
        },
        "regime_type": {
          "type": "string",
          "enum": [
            "persistent_positive",
            "persistent_negative",
            "transitional",
            "low_conviction"
          ]
        },
        "positive_days": {
          "type": "integer"
        },
        "negative_days": {
          "type": "integer"
        },
        "avg_magnitude_billions": {
          "type": "number"
        },
        "sign_flips": {
          "type": "integer"
        },
        "persistence_pct": {
          "type": "number"
        },
        "confidence": {
          "type": "integer"
        },
        "reasoning": {
          "type": "string"
        }
      },
      "required": [
        "regime_detected",
        "regime_type",
        "positive_days",
        "negative_days",
        "avg_magnitude_billions",
        "sign_flips",
        "persistence_pct",
        "confidence",
        "reasoning"
      ]
    }
  }
}
```

Appendix A.4. Request Body

Each line of a batch input file is one request of the following form.

```
{
  "custom_id": "<arm>|<symbol>|<window_end>",
  "method": "POST",
  "url": "/v1/chat/completions",
  "body": {
    "model": "gpt-6-luna",
    "messages": [
      {
        "role": "user",
        "content": "<prompt text of Appendix A.2>"
      }
    ],
```

```
    "max_completion_tokens": 4000,
    "reasoning_effort": "medium",
    "response_format": {
      "type": "json_schema",
      "json_schema": "<schema of Appendix A.3>"
    }
  }
}
```

Appendix B. Per-Year Tables

Tables A1–A3 give the numbers behind the figures of Section 4 for every year. The years are the calendar years of the window end, and rates are percentages of parsed windows. The Wilson intervals treat overlapping windows as independent and are optimistic; the quarter-block bootstrap interval of every rate is in the result files of the repository (Appendix C).

Table A1. Mean absolute daily GEX (USD billion) by series and year, SPY. Each mean is taken over the days that the series has in common with `new_full`; the `new_full` column and the day count use the days shared with `new_ex0dte`. The means of the complete series in Section 4.1 are taken over the days it shares with the published inputs, so they can differ slightly from the `new_full` column ($6.48 against $6.46 billion in 2024). A dash marks a year outside the series.

Year	Days	jan2026_phase4a	call_only	old_bs	new_ex0dte	new_full
2008	253	–	–	1.39	1.59	1.60
2009	252	–	–	0.98	0.88	0.88
2010	252	–	–	1.01	1.14	1.17
2011	252	–	–	1.79	2.00	2.03
2012	250	–	–	1.54	1.71	1.78
2013	252	–	–	2.34	2.29	2.51
2014	252	–	–	3.03	3.07	3.36
2015	252	–	–	2.69	2.91	3.37
2016	252	–	–	3.99	3.76	4.25
2017	251	–	–	6.43	5.71	6.38
2018	250	–	–	4.79	4.63	5.32
2019	251	–	–	3.62	3.39	4.11
2020	252	3.24	10.51	2.76	2.57	3.24
2021	252	4.97	17.13	3.98	3.50	4.99
2022	251	5.48	9.27	4.42	4.62	5.46
2023	250	10.62	17.09	8.39	4.59	6.10
2024	252	20.19	20.19	16.19	4.13	6.46
2025	250	19.21	19.21	14.71	4.55	7.13
2026	194	–	21.68	6.00	5.77	8.38

Table A2. Detection rate of the deterministic classifier by series and window-end year, SPY, in percent of windows. Windows are the windows of `new_full`. A dash marks a year without windows in the series.

Year	Windows	`jan2026_phase4a`	`call_only`	`old_bs`	`new_ex0dte`	`new_full`
2008	224	–	–	0.0	0.0	0.0
2009	252	–	–	0.0	0.0	0.0
2010	252	–	–	0.0	0.0	0.0
2011	252	–	–	0.0	0.0	0.0
2012	250	–	–	0.0	0.0	0.0
2013	252	–	–	0.0	0.0	0.0
2014	252	–	–	13.5	8.3	9.1
2015	252	–	–	0.4	1.6	7.9
2016	252	–	–	26.4	22.2	28.6
2017	251	–	–	38.8	17.9	30.7
2018	250	–	–	32.1	24.0	34.8
2019	251	–	–	7.2	3.2	10.0
2020	252	12.1	90.6	16.7	10.7	22.2
2021	252	4.0	100.0	3.6	0.0	3.6
2022	251	32.3	100.0	26.3	36.7	40.6
2023	250	19.3	100.0	22.5	25.2	22.4
2024	252	100.0	100.0	100.0	13.1	10.3
2025	250	100.0	100.0	100.0	16.4	13.6
2026	194	–	100.0	23.5	21.1	11.3

Table A3. Detection rate of gpt-6-luna by arm and window-end year, in percent of parsed windows (a dash marks a year without windows in the arm). The columns are the arms `full_spy`, `ex0dte_spy`, `full_qqq`, `full_qqq_scaled`, `ex0dte_qqq`, `full_iwm`, `full_iwm_scaled` and `ex0dte_iwm` of Table 3 (full = `new_full`, scaled = `new_full_scaled`, ex-0DTE = `new_ex0dte`); the bridge, intraday and Phase 3/4 arms are reported in Sections 4.3, 4.9 and 4.11.

	SPY		QQQ			IWM		
Year	full	ex-0DTE	full	scaled	ex-0DTE	full	scaled	ex-0DTE
2008	0.0	0.0	–	–	–	0.0	–	–
2009	0.0	0.0	–	–	–	0.0	–	–
2010	0.0	0.0	–	–	–	0.0	–	–
2011	0.0	0.0	0.0	–	–	0.0	–	–
2012	0.0	0.0	0.0	–	–	0.0	–	–
2013	0.0	0.0	0.0	–	–	0.0	–	–
2014	9.1	8.3	0.0	–	–	0.0	–	–
2015	7.9	1.6	0.0	–	–	0.0	–	–
2016	28.2	22.2	0.0	–	–	0.0	–	–
2017	30.7	17.9	0.0	–	–	0.0	–	–
2018	35.2	23.2	0.0	–	–	0.0	–	–
2019	10.0	3.2	0.0	–	–	0.0	–	–
2020	22.2	10.7	0.0	0.0	0.0	0.0	9.9	0.0
2021	3.6	0.0	0.0	23.8	0.0	0.0	65.9	0.0
2022	41.8	37.1	0.0	61.0	0.0	0.0	100.0	0.0
2023	23.2	26.0	0.0	17.6	0.0	0.0	79.2	0.0
2024	11.1	14.3	0.0	8.4	0.0	0.0	20.6	0.0
2025	14.0	16.4	0.0	20.4	0.0	0.0	51.2	0.0
2026	11.3	21.1	0.0	16.5	0.0	0.0	45.4	0.0

Appendix C. Data Availability Details

The repository holds the condensed material and the results. The condensed files are in `data/backups/gex_llm/condensed/`.

- `detections.csv.gz`: 49,090 rows, one per stored model answer, with the verdict, the reported metrics and the parse status but without the reasoning or the raw text. The key is `arm`, `series_key`, `symbol`, `window_end`, `model`, `prompt_version`.
- `windows.csv.gz`: 38,592 rows, one per 30-day window, with the labels, the sha256 of the input as sent and the verdict of the deterministic rule but without the sequences.
- `series_index.csv`: 16 rows, one per series and symbol, with the row count, the first and last date, the source table and the hash of the builder.
- `runs.csv.gz`: one row per Batch job, local run, pilot or backfilled run, with the batch and file identifiers or the engine record, the sha256 of the input and output files, the token counts and the cost.
- `run_records.jsonl.gz`: the manifest and the batch record of every run folder (`manifest.json` and `batch_<k>.json`), one line per file.

Every detection rate and agreement figure can be recomputed from the first two files (join on `series_key`, `symbol` and `window_end`); the other numbers of the article are in the files of `docs/papers/electronics/results/`, and the daily values behind them are in the full tables. The provenance of every Batch job and local run is in the last two files, which are written from the per-run files (`tables/runs/`, and `manifest.json` and `batch_<k>.json` in each run folder). The per-run files, the raw request and answer files (`input_<k>.jsonl`, `output_<k>.jsonl`) and the full tables are kept on the authors' machine and in the database; they are not part of the repository, and the full tables are deposited at Zenodo.

The full tables are loaded into four tables and one view of the schema `backups` of the research database: `gex_llm_regime_series` (the number shown to the model, one row per series, symbol and trading date), `gex_llm_regime_windows` (the 30-day input as sent, its sha256 and the verdict of the rule), `gex_llm_regime_runs` (one row per Batch job, local run, pilot or backfilled run) and `gex_llm_regime_detections` (one row per model answer), with the view `gex_llm_regime_rates_summary` (yearly rate of the model next to the rate of the rule, by arm). The parquet copies of these four tables are deposited at Zenodo (Data Availability Statement). The options data are subject to the vendor's licence and are not redistributed.

Supplementary Materials: The following supporting information is available with this article: Section S1, series definitions in SQL; Tables S1–S4, run ledger.

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Data Availability Statement: All code, the condensed per-window tables and the result files are available at https://github.com/iAmGiG/gex-llm-patterns (accessed on 2 November 2026). The full tables (daily series, windows, runs and detections) are deposited at Zenodo; the DOI is added at submission. The options data were obtained from Alpha Vantage under a premium API subscription whose terms permit this use; the raw option chains are not part of the deposit, which holds the daily aggregates shown to the models.

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Abbreviations

The following abbreviations are used in this manuscript:

0DTE	Zero Days to Expiration
API	Application Programming Interface
EOD	End of day (the daily closing snapshot)
ETF	Exchange-Traded Fund
GEX	Gamma Exposure
GPU	Graphics Processing Unit
ISO	International Organization for Standardization
LLM	Large Language Model
OI	Open Interest
SQL	Structured Query Language

References

1. Regan, C.; Xie, Y. Validating LLM Structural Reasoning: Detecting Persistent Market Regimes Through Temporal Obfuscation. Artificial Intelligence Applications and Innovations (AIAI 2026), 22nd IFIP WG 12.5 International Conference; Springer: Cham, 2026; IFIP Advances in Information and Communication Technology, pp. 127–140. doi:10.1007/978-3-032-30801-6_10.
2. Regan, C.; Xie, Y. Inferring Latent Market Forces: Evaluating LLM Detection of Gamma Exposure Patterns via Obfuscation Testing. 2nd IEEE International Workshop on Large Language Models for Finance (LLM-Finance), IEEE International Conference on Big Data (BigData); IEEE: Macau, China, 2025. arXiv:2512.17923.
3. Regan, C.; Xie, Y. Temporal Obfuscation Testing for LLM Structural Reasoning: From Single-Day Dealer Constraints to Persistent Market Regimes. *Journal of Risk and Financial Management* **2026**, *19*, 382. doi:10.3390/jrfm19060382.
4. Grossman, S.J.; Miller, M.H. Liquidity and Market Structure. *Journal of Finance* **1988**, *43*, 617–633. doi:10.1111/j.1540-6261.1988.tb04594.x.
5. Frey, R. Derivative Asset Analysis in Models with Level-Dependent and Stochastic Volatility. *CWI Quarterly* **1997**, *10*, 1–34.
6. Anderegg, B.; Ulmann, F.; Sornette, D. The Impact of Option Hedging on the Spot Market Volatility. *Journal of International Money and Finance* **2022**, *124*, 102627. doi:10.1016/j.jimonfin.2022.102627.
7. Ni, S.X.; Pearson, N.D.; Poteshman, A.M. Stock Price Clustering on Option Expiration Dates. *Journal of Financial Economics* **2005**, *78*, 49–87. doi:10.1016/j.jfineco.2004.08.005.
8. Gârleanu, N.; Pedersen, L.H.; Poteshman, A.M. Demand-Based Option Pricing. *Review of Financial Studies* **2009**, *22*, 4259–4299. doi:10.1093/rfs/hhp005.
9. Baltussen, G.; Da, Z.; Lammers, S.; Martens, M. Hedging Demand and Market Intraday Momentum. *Journal of Financial Economics* **2021**, *142*, 377–403. doi:10.1016/j.jfineco.2021.04.029.
10. SpotGamma. Understanding Gamma Exposure. Technical Documentation, 2021.
11. Fishman, R. All You Ever Wanted To Know About Gamma, Op-Ex, And Option-Driven Equity Flows. Technical report, Goldman Sachs Equity Derivatives Strategy, 2023. Republished by SpotGamma.
12. CBOE Global Markets. Zero Days to Expiration Options (0DTE): Market Structure and Trading Activity. CBOE Research Report, 2024.
13. CBOE Global Markets. SPX 0DTE Options Jump to Record 62% Share in August. CBOE Insights, 2025.
14. Dim, C.; Eraker, B.; Vilkov, G. 0DTEs: Trading, Gamma Risk and Volatility Propagation. *SSRN Working Paper* **2024**. SSRN id 4692190, doi:10.2139/ssrn.4692190.
15. Adams, G.; Dim, C.; Eraker, B.; Fontaine, J.S.; Ornthanalai, C.; Vilkov, G. Do S&P500 Options Increase Market Volatility? Evidence from 0DTEs. *SSRN Working Paper* **2025**. SSRN id 5641974.
16. Hamilton, J.D. A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle. *Econometrica* **1989**, *57*, 357–384. doi:10.2307/1912559.
17. Ang, A.; Bekaert, G. International Asset Allocation with Regime Shifts. *Review of Financial Studies* **2002**, *15*, 1137–1187. doi:10.1093/rfs/15.4.1137.

18. Nystrup, P.; Madsen, H.; Lindström, E. Dynamic Portfolio Optimization across Hidden Market Regimes. *Quantitative Finance* **2018**, *18*, 83–95. doi:10.1080/14697688.2017.1342857.
19. Lopez-Lira, A.; Tang, Y. Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models. arXiv preprint arXiv:2304.07619, 2023. Preprint.
20. Xiao, Y.; Sun, E.; Luo, D.; Wang, W. TradingAgents: Multi-Agents LLM Financial Trading Framework. arXiv preprint arXiv:2412.20138, 2024. Preprint.
21. Wei, J.; Wang, X.; Schuurmans, D.; Bosma, M.; Ichter, B.; Xia, F.; Chi, E.; Le, Q.; Zhou, D. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. *Advances in Neural Information Processing Systems* **2022**, *35*, 24824–24837.
22. Kojima, T.; Gu, S.S.; Reid, M.; Matsuo, Y.; Iwasawa, Y. Large Language Models are Zero-Shot Reasoners. *Advances in Neural Information Processing Systems* **2022**, *35*, 22199–22213.
23. McCoy, R.T.; Yao, S.; Friedman, D.; Hardy, M.; Griffiths, T.L. Embers of Autoregression Show How Large Language Models Are Shaped by the Problem They Are Trained to Solve. *Proceedings of the National Academy of Sciences* **2024**, *121*, e2322420121. doi:10.1073/pnas.2322420121.
24. Lopez-Lira, A.; Tang, Y.; Zhu, M. The Memorization Problem: Can We Trust LLMs' Economic Forecasts? arXiv preprint arXiv:2504.14765, 2025. Preprint.
25. Dong, Y.; Jiang, X.; Liu, H.; Jin, Z.; Gu, B.; Yang, M.; Li, G. Generalization or Memorization: Data Contamination and Trustworthy Evaluation for Large Language Models. Findings of the Association for Computational Linguistics: ACL 2024; Association for Computational Linguistics: Bangkok, Thailand, 2024; pp. 12039–12050.
26. Ribeiro, M.T.; Wu, T.; Guestrin, C.; Singh, S. Beyond Accuracy: Behavioral Testing of NLP Models with CheckList. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 2020, pp. 4902–4912. doi:10.18653/v1/2020.acl-main.442.
27. Cronbach, L.J.; Meehl, P.E. Construct Validity in Psychological Tests. *Psychological Bulletin* **1955**, *52*, 281–302.
28. Black, F.; Scholes, M. The Pricing of Options and Corporate Liabilities. *Journal of Political Economy* **1973**, *81*, 637–654. doi:10.1086/260062.
29. OpenAI. gpt-oss-120b & gpt-oss-20b Model Card. arXiv preprint arXiv:2508.10925, 2025.
30. Kwon, W.; Li, Z.; Zhuang, S.; Sheng, Y.; Zheng, L.; Yu, C.H.; Gonzalez, J.E.; Zhang, H.; Stoica, I. Efficient Memory Management for Large Language Model Serving with PagedAttention. Proceedings of the 29th Symposium on Operating Systems Principles (SOSP '23); ACM: Koblenz, Germany, 2023; pp. 611–626. doi:10.1145/3600006.3613165.
31. Grattafiori, A.; Dubey, A.; Jauhri, A.; others. The Llama 3 Herd of Models. arXiv preprint arXiv:2407.21783, 2024. doi:10.48550/arXiv.2407.21783.
32. Yang, A.; Li, A.; Yang, B.; others. Qwen3 Technical Report. arXiv preprint arXiv:2505.09388, 2025. doi:10.48550/arXiv.2505.09388.
33. Brown, L.D.; Cai, T.T.; DasGupta, A. Interval Estimation for a Binomial Proportion. *Statistical Science* **2001**, *16*, 101–133. doi:10.1214/ss/1009213286.
34. Newcombe, R.G. Interval Estimation for the Difference Between Independent Proportions: Comparison of Eleven Methods. *Statistics in Medicine* **1998**, *17*, 873–890. doi:10.1002/(SICI)1097-0258(19980430)17:8<873::AID-SIM779>3.0.CO;2-I.
35. Künsch, H.R. The Jackknife and the Bootstrap for General Stationary Observations. *The Annals of Statistics* **1989**, *17*, 1217–1241. doi:10.1214/aos/1176347265.
36. Cohen, J. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* **1960**, *20*, 37–46. doi:10.1177/001316446002000104.
37. McNemar, Q. Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages. *Psychometrika* **1947**, *12*, 153–157. doi:10.1007/BF02295996.

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